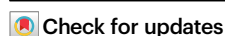


Distinctly different active sites of ZnO-ZrO₂ catalysts in CO₂ and CO hydrogenation to methanol reactions

Received: 20 December 2024

Accepted: 9 May 2025

Published online: 18 May 2025



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The active site of a solid catalyst varies sensitively with the catalyzed reaction. Herein, using experimentally measured elementary surface reaction kinetics of CO₂ or CO hydrogenation reactions over a ZnO-ZrO₂ catalyst under working conditions in combinations with comprehensive structural characterizations and theoretical simulations, we unveil the distinctly different active sites in catalyzing the CO₂ or CO hydrogenation to methanol reaction. Zn²⁺ cations with different local environments are present on the ZnO-ZrO₂ surface, including Zn₁ single atoms exclusively with a Zn-O-Zr local structure and Zn_n clusters with both Zn-O-Zr and Zn-O-Zn local structures. The -Zr-O-Zr- structure bonded to the Zn_n clusters is more easily to be reduced than that bonded to the Zn₁ single atoms. The Zn₁-single atom (-Zr-O-Zn-O-Zr-) is the active site for catalyzing the CO₂ hydrogenation to methanol reaction, whereas the Zn_n cluster bonded to an in situ formed -Zr-V_o-Zr- structure (-Zn-O-Zn(-O-Zr-V_o-Zr)-O-Zr-) is the active site for catalyzing the CO hydrogenation to methanol reaction. These results provide a reliable and effective methodology of elementary surface reaction kinetics for identifications of active sites of working catalysts in complex reactions and unveil how sensitively the active site structure varies with the catalyzed reaction.

Since the postulation of the concept of active site in 1925¹, delineating active sites in solid catalysts has become the central pursuit in the fundamental studies of heterogeneous catalysis² but meanwhile remained as an enduring challenge due to its inherent complexity^{3–7}. The active site of a solid catalyst sensitively varies with the catalyzed reaction. Cu-ZnO based catalysts for CO_x hydrogenation to methanol reactions are a well-known example. After long-

term studies and strong arguments, the in situ formed CuZn alloy^{8–11} or Cu(I)_{Cu}-hydroxylated ZnO interface^{12–19} were demonstrated by experimental evidence as the active structures of working Cu-ZnO based catalysts in the CO or CO₂ hydrogenation to methanol reactions, respectively. ZnO-ZrO₂ binary oxides have recently emerged as novel catalysts for the CO₂ hydrogenation to methanol reaction^{20,21}, and in combination with various types of zeolites, as highly active

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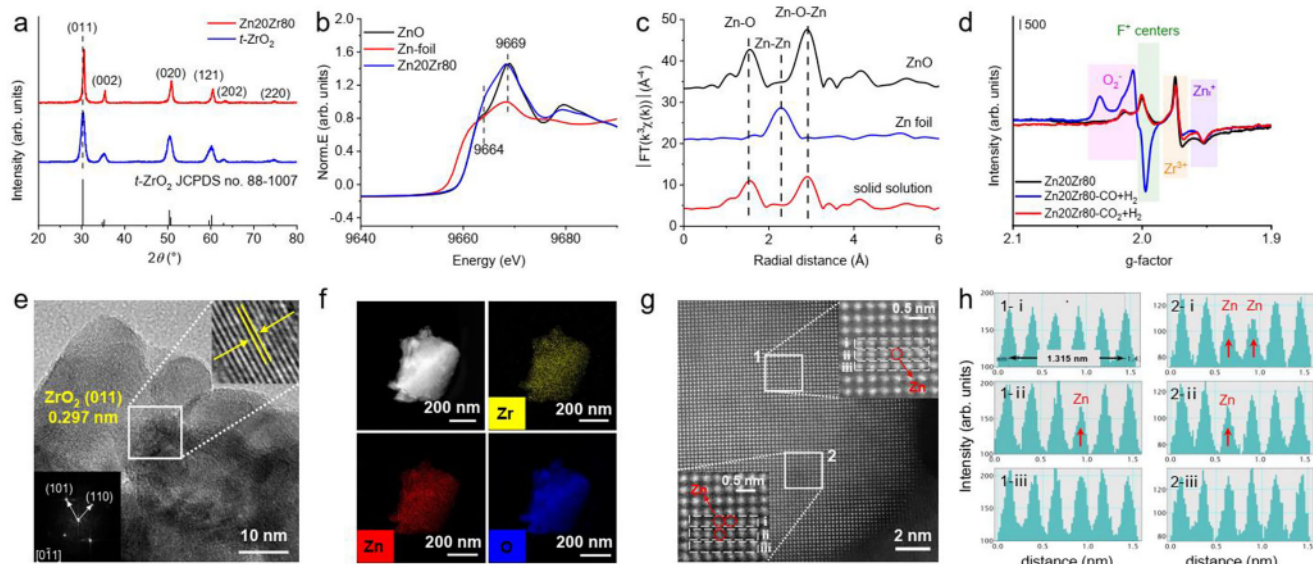


Fig. 1 | Structural characterizations of Zn20Zr80 catalyst. **a** X-Ray diffraction pattern of Zn20Zr80 and *t*-ZrO₂. **b** Zn K-edge XANES spectra and **c** Fourier transforms of the k^3 -weighted Zn K-edge EXAFS spectra of Zn20Zr80 catalyst and referring ZnO and Zn foil. **d** ESR spectra of Zn20Zr80 calcined in Ar at 773 K, and subsequently treated under 3 MPa CO + H₂ (H₂:CO = 2) or 3 MPa CO₂ + H₂ (H₂:CO₂ = 3) at 573 K without exposure to air. All the spectra were measured at 140 K. **e** representative HRTEM image of Zn20Zr80 catalyst with an inserted electron

diffraction pattern. **f** HAADF-STEM image and corresponding EDS mapping images of Zn20Zr80 catalyst. **g** Atomically resolved STEM images of edge areas of Zn20Zr80 nanoparticle, and **h** single-pixel line profiles across the atom rows of 1-i, 1-ii, 1-iii and 2-i, 2-ii, 2-iii within rectangles 1 and 2 marked in (**g**). The dark dots in 1g and low-intensity single-pixel line profiles in 1h correspond to Zn-containing atomic columns, marked with red circles and arrows, respectively.

bifunctional catalysts for the CO_x hydrogenation to value-added chemicals with methanol as the reaction intermediate^{22–26}, but the related active sites and reaction mechanisms are elusive. Different types of active structures were proposed in the CO₂^{27–30} or CO^{24,31} hydrogenation to methanol reactions. Meanwhile, although the formate (HCOO*) pathway is commonly accepted in the CO_x hydrogenation to methanol reaction^{20,27,30}, different types of formate species, bridging-adsorbed formate species (bri-HCOO*)³² and tri-HCOO* species in a tetrahedral configuration³³, were argued as the active surface formate intermediate. Theoretical calculations also proposed a sequential hydrogenation mechanism of CO to methanol via HCO*, H₂CO*, H₃CO* intermediates³¹. Previous mechanistic studies of ZnO-ZrO₂ catalyzed CO_x hydrogenation reactions were mostly conducted at near atmospheric pressure^{20,29,30,32,34} and occasionally at around 1.5 MPa^{35,36}, which are quite milder than the working conditions, whereas structures of oxide catalysts under the reductive CO_x hydrogenation reactions sensitively varied with the reaction pressure and temperature³⁷. Thus, it is necessary to fundamentally investigate ZnO-ZrO₂ catalyzed CO_x hydrogenation reactions under the working conditions. Recently, elementary surface reaction kinetics of the formate hydrogenation reaction were successfully acquired on working oxide catalysts in the CO³⁸ or CO₂³⁹ hydrogenation to methanol reactions using temporal in situ DRIFTS spectroscopy, which, compared to the corresponding macroscopic reaction kinetics, unambiguously identified the active formate species.

In this work, via experimental studies of elementary surface reaction kinetics of CO_x hydrogenation reaction over a ZnO-ZrO₂ catalyst with an atomic Zn:Zr ratio of 1:4 (denoted as Zn20Zr80) under working conditions in combination with comprehensive structural characterizations and theoretical calculations, we unambiguously identify the Zn₁-single atom (-Zr-O-Zn-O-Zr-O-Zr-) and the partially-reduced Zn_n cluster with in situ formed -Zr-V_o-Zr- structure (-Zn-O-Zn(-O-Zr-V_o-Zr)-O-Zr-O-Zr-) as the active sites for catalyzing the CO₂ and CO hydrogenation to methanol reactions, respectively. The active formate species and associate reaction mechanisms are also identified.

Results

Structural characterizations

A Zn20Zr80 catalyst synthesized via a co-precipitation method using Zr(NO₃)₄·5H₂O and Zn(NO₃)₂·6H₂O as the precursors⁴⁰ exhibits a Zn content of 20.3 at.% and a BET specific surface area of 23.1 m²/g. Its X-Ray diffraction (XRD) pattern (Fig. 1a) matches that of the standard pattern of pure tetragonal ZrO₂ phase (*t*-ZrO₂) (JCPDS file no. 88-1007), but the (011) spacing shifts to a higher angle than that of a reference pure *t*-ZrO₂ sample. This is an indication of substitutions of lattice Zr⁴⁺ cations by smaller-sized Zn²⁺ cations in Zn20Zr80. In the Zn K-edge X-ray absorption near-edge structure (XANES) spectra (Fig. 1b), Zn20Zr80 shows an almost identical peak at 9669 eV to pure ZnO but a stronger pre-edge shoulder peak at 9664 eV, which arise from to electron transitions from Zn 1s to Zn 4p-O 2p and Zn 4sp-O 2p hybridized states of the conduction band⁴¹, respectively. The pre-edge feature of Zn K-edge XANES of ZnO was previously reported to increase as the ZnO particle size decreased^{42,43}. In the corresponding Zn K-edge extended X-ray absorption fine structure Fourier transforms (FT-EXAFS) spectra (Fig. 1c, Supplementary Fig. 1, Supplementary Table 1), Zn20Zr80 shows the nearest neighboring Zn-O and Zn-Zn coordination shells of ZnO at approximately 1.8 Å (coordination number = 4) and 3.2 Å (coordination number = 12.2)²⁹, respectively. These Zn K-edge XAS results suggest the presence of ultrafine ZnO particles in Zn20Zr80 which are invisible in the XRD pattern. Thus, Zn20Zr80 consists of Zn²⁺-substituted *t*-ZrO₂ solid solution and fine ZnO particles, consistent with previous reports^{29,34,35,44–46}. Electron paramagnetic resonance (ESR) spectra of Zn20Zr80 calcined in Ar at 773 K without exposure to air (Fig. 1d and Supplementary Fig. 2) only show signals of O₂⁻, F⁺ centers, Zr³⁺ and interstitial Zn⁺ defects. The inverse EPR susceptibility χ_{EPR} of the observed ESR signals measured at different temperatures (Supplementary Fig. 3) were found to follow the Curie-Weiss law^{47,48}, confirming that these ESR signals arise from the pragmatically isolated sites. The observed ESR features barely change after an exposure of calcined Zn20Zr80 to air (Supplementary Fig. 4), suggesting that the F⁺ centers, Zr³⁺ and interstitial Zn⁺ defects locate in the bulk. X-ray

Photoelectron Spectroscopy (XPS) spectrum (Supplementary Fig. 5) shows Zn^{2+} and Zr^{4+} cations on the Zn20Zr80 surface.

TEM images of Zn20Zr80 (Supplementary Fig. 6) show large aggregates consisting of fine nanoparticles, while careful analysis of HRTEM images (Fig. 1e and Supplementary Fig. 7) only gives the lattice fringes of $t\text{-ZrO}_2$ but not of ZnO, suggesting that the ultrafine ZnO particles probably embed in the bulk of Zn20Zr80 inaccessible by HRTEM characterizations. Low-resolution HAADF-STEM and corresponding elemental mapping images (Fig. 1f) demonstrate rather uniform distributions of Zr and Zn within Zn20Zr80, but examinations of the edge areas with atomically resolved STEM images and corresponding EDS mapping images (Fig. 1g and Supplementary Fig. 8) unveil an uneven Zn distribution on the Zn20Zr80 surface. The corresponding single-pixel line profiles across the atom rows in the inset of Fig. 1g (Fig. 1h) show columns with an average spacing of 0.260 nm corresponding to the {002} crystal planes of $t\text{-ZrO}_2$. The dominant lattice atoms are the bright Zr atoms, while a few lattice atoms are the dark Zn atoms. This directly visualizes the substitutions of lattice Zr^{4+} cation by Zn^{2+} cations. Moreover, such an analysis demonstrates the presence of Zn_1 single atom and Zn_3 cluster in the area 1 and 2, respectively. Similarly, the Zn_2 and Zn_4 clusters were also identified (Supplementary Fig. 9). Thus, the Zn20Zr80 surface is the Zn^{2+} -substituted $t\text{-ZrO}_2$ solid solution with the Zn^{2+} cations of different local environments, including the Zn_1 single atom exclusively with the Zn-O-Zr local structure and the Zn_n clusters with both the Zn-O-Zr and Zn-O-Zn local structures.

CH_3OH adsorption was used to further probe the surface structure of Zn20Zr80 (Supplementary Fig. 10). Vibrational bands at 1156, 1060/1045, and 2871/1598/1374 cm^{-1} appear upon CH_3OH adsorption on Zn20Zr80 at RT, which, compared to those on the reference $t\text{-ZrO}_2$ and ZnO samples, can be assigned to the methoxy group at the Zr^{4+} site ($\text{CH}_3\text{O}_{\text{Zr}}^*$), methoxy group at the Zn^{2+} sites with different local environments ($\text{CH}_3\text{O}_{\text{Zn}}^*$), and bri-HCOO^* species at the Zn^{2+} sites, respectively. CH_3OH dissociation is more extensive at the Zn^{2+} site on the Zn20Zr80 surface than at the Zr^{4+} site. Meanwhile, CH_3OH dissociation is more extensive at the Zn^{2+} site on the Zn20Zr80 surface than on the ZnO surface, but less extensive at the Zr^{4+} site on the Zn20Zr80 surface than on the $t\text{-ZrO}_2$ surface. These also support that the Zn20Zr80 surface is the Zn^{2+} -substituted $t\text{-ZrO}_2$ solid solution.

Catalytic performance in CO_x hydrogenation reaction

Catalytic performance of Zn20Zr80 in CO_2 or CO hydrogenation reaction was evaluated. As the reaction temperature increasing from 523 to 573 K in the CO_2 hydrogenation reaction, the CO_2 conversion increases from 1.7% to 9.1% while the CH_3OH selectivity decreases from 89.7 to 80.3% (Fig. 2a). The CO_2 conversion of 9.1% at 573 K is close to the equilibrium value (around 9.3%) under the adopted reaction condition⁴⁹. We then measured the catalytic performance within the kinetics-controlled range (Supplementary Table 2). The derived Arrhenius plots (Fig. 2b and Supplementary Fig. 11) calculated from the CH_3OH formation and CO_2 reaction rates give similar apparent activation energy (E_a) of 81.9 ± 1 and $85.1 \pm 1 \text{ kJ mol}^{-1}$, respectively,

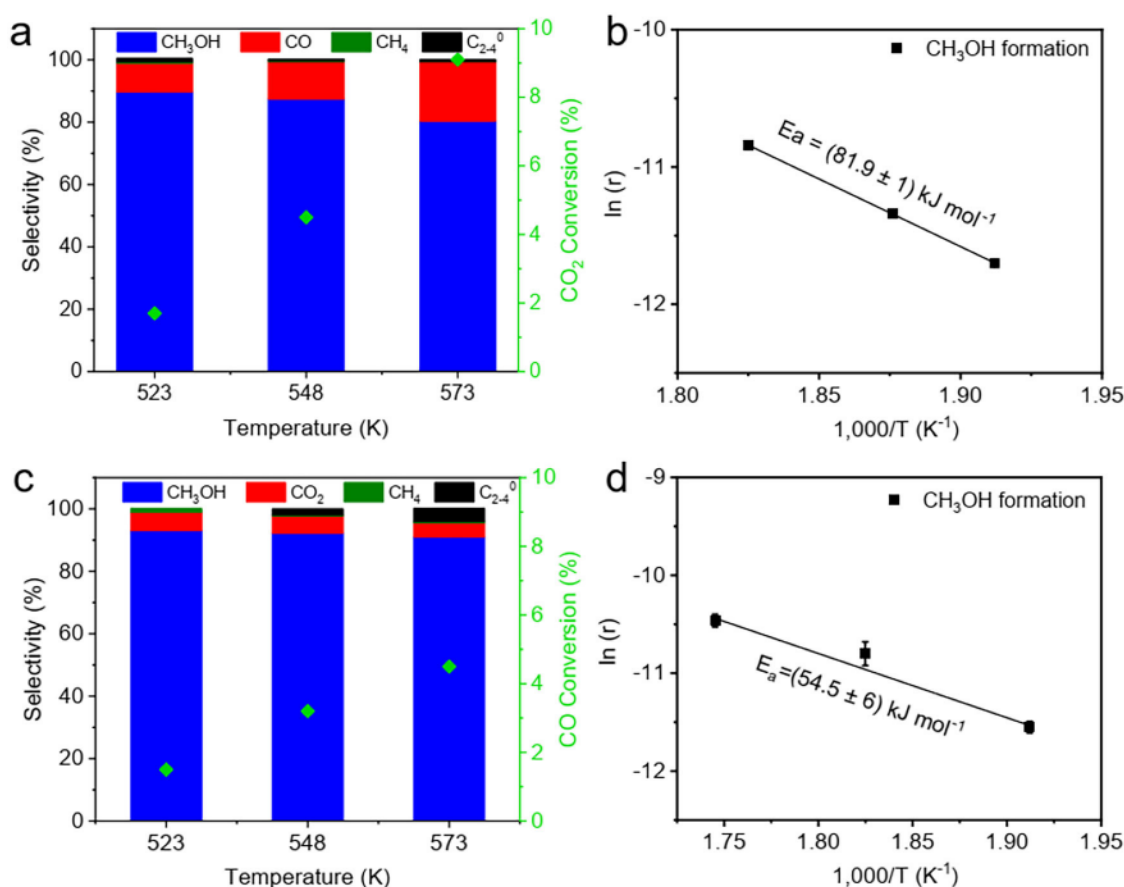


Fig. 2 | Catalytic performance of Zn20Zr80 catalyst. **a** Catalytic performance of Zn20Zr80 catalyst in CO_2 hydrogenation reaction (reaction condition: H_2 : CO_2 = 3, 3 MPa; flow rate: 30 mL min^{-1} ; catalyst mass: 1 g) and **b** Arrhenius plots of CO_2 hydrogenation to methanol reaction catalyzed by Zn20Zr80 catalyst derived from the data summarized in Supplementary Table 2 (reaction condition: H_2 : CO_2 = 3,

3 MPa; flow rate: 30 mL min^{-1} ; catalyst mass: 600 mg catalyst diluted with 400 mg SiC). **c** Catalytic performance of Zn20Zr80 catalyst in CO hydrogenation reaction (reaction condition: H_2 : CO = 2, 3 MPa; flow rate: 30 mL min^{-1} ; catalyst mass: 1 g) and **d** derived Arrhenius plots of CO hydrogenation to methanol reaction. The error bars in the figure represent the standard errors (SE) of the fitted values.

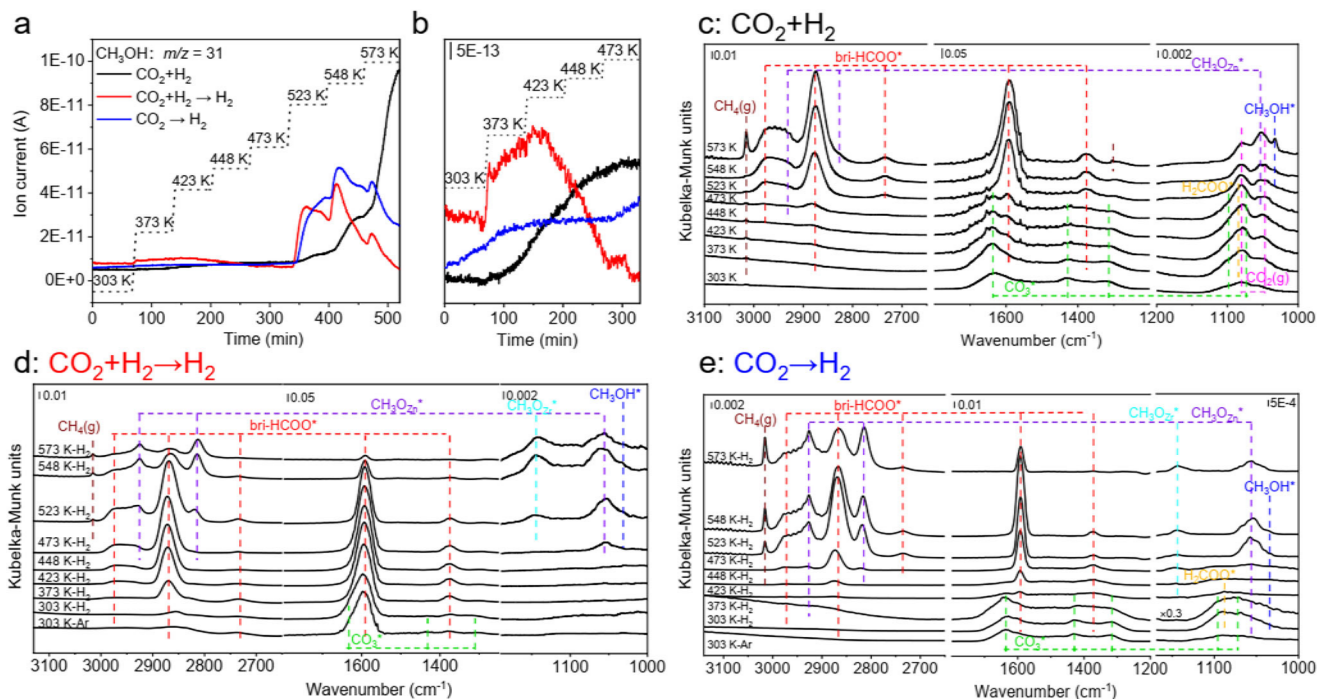


Fig. 3 | Operando characterizations of CO₂ hydrogenation reaction. **a** CH₃OH MS signals as a function of temperature over Zn₂₀Zr₈₀ under 3 MPa CO₂ + H₂ (H₂: CO₂ = 3) (CO₂ + H₂), Zn₂₀Zr₈₀ pretreated in 3 MPa CO₂ + H₂ at 573 K for 60 min, cooled to 303 K and purged in Ar under 3 MPa H₂ (CO₂ + H₂ → H₂), and Zn₂₀Zr₈₀

pretreated in 3 MPa CO₂ at 303 K for 60 min and purged in Ar under 3 MPa H₂ (CO₂ → H₂) and **b**, zoom-in curves between 303 and 473 K. Corresponding steady-state in situ DRIFTS spectra at indicated temperatures of **c**, CO₂ + H₂, **d**, CO₂ + H₂ → H₂ and **e**, CO₂ → H₂.

consistent with the literature results^{50,51}. The apparent activation energy for CO production was also calculated as 125.9 ± 7 kJ mol⁻¹, much higher than that of CH₃OH formation. Thus, the mechanism of reverse water gas shift reaction followed by CO hydrogenation should barely contribute to the CH₃OH production by CO₂ hydrogenation reaction in our case, although it could not be fully excluded at high CO₂ conversions.

For the CO hydrogenation reaction, the CO conversion increases from 1.5 to 4.5% as the reaction temperature increasing from 523 to 573 K, while the CH₃OH selectivity decreases from 93.0% to 91.0% (Fig. 2c). The corresponding Arrhenius plots (Fig. 2d and supplementary Fig. 11) using CH₃OH formation and CO consumption rate give E_a of 54.5 ± 6 and 54.3 ± 6 kJ mol⁻¹, respectively. The apparent activation energy for CO₂ production was also calculated as 37.3 ± 8 kJ mol⁻¹. Considering the low CO conversions and very low CO₂ selectivity in our case, the reaction pathway of CO₂ production and its consequent hydrogenation to methanol can be ignored in the CO hydrogenation to methanol reaction.

Reaction mechanism of CO₂ hydrogenation reaction

ESR spectrum of Zn₂₀Zr₈₀ subjected to the CO₂ hydrogenation reaction at 573 K without exposure to air is almost identical to that calcined in Ar at 773 K (Fig. 1d), indicating that Zn₂₀Zr₈₀ is not reduced during the reaction. Additional oxygen vacancies were observed on ZnZrO_x and spinel oxides under an H₂ atmosphere and can be quenched by a CO₂ environment^{34,52}, but few oxygen vacancies were generated on ZnZrO_x in a CO₂ + H₂ atmosphere³⁶, consistent with our observations. Zn₂₀Zr₈₀-catalyzed CO₂ hydrogenation reaction was operando characterized via a combination of in situ DRIFTS and online mass spectroscopy mounted after the decompression valve of the in situ high temperature and high-pressure reactor cell for DRIFTS measurements. As shown in Fig. 3a, b, very weak gaseous CH₃OH signals was detected at a reaction temperature as low as 373 K, and the intensity increased with the reaction temperatures slowly up to 473 K

and then rapidly. Figure 3c and Supplementary Fig. 12 show the corresponding in situ DRIFTS spectra, and Supplementary Table 3 summarizes assignments of observed vibrational bands. Adsorbed CO₃* species dominates on the surface at 303 K. With the reaction temperature increasing, CO₃* slightly strengthens at 373 K, then keeps weakening, and disappears at 523 K; adsorbed H₂COO* species emerges at 373 K and keeps growing up to 448 K, then weakens and disappears at 523 K; adsorbed bri-HCOO* species emerges at 373 K and keeps growing up to 573 K; adsorbed CH₃O* species at the Zn site (CH₃O_{Zn}*) appears at 473 K and keeps growing up to 573 K; adsorbed methanol species (CH₃OH*) appears at 523 K and keeps growing up to 573 K.

A Zn₂₀Zr₈₀ catalyst pretreated in CO₂ hydrogenation at 573 K for 1 h, cooled to 303 K and purged in Ar exhibits the bri-HCOO*, CO₃* and CH₃OH* species whose hydrogenation reactivity was studied in 3 MPa H₂. Weak gaseous CH₃OH signals emerge as soon as the catalyst is heated, increases slowly but then decreases and disappears at 473 K, while strong CH₃OH productions occur again at 523 K and above (Fig. 3a, b). As shown in the corresponding in-situ DRIFTS spectra (Fig. 3d and Supplementary Fig. 13), CH₃OH* desorbs at low temperatures, giving the low-temperature CH₃OH productions; CO₃* disappears at 373 K while bri-HCOO* grows up to 473 K and then weakens up to 573 K; CH₃O_{Zn}* emerges at 473 K, grows up to 548 K and then weakens at 573 K; both CH₃O_{Zn}* and CH₃OH* emerge at 523 K, grow at 548 K and weakens at 573 K. These observations demonstrate the existence of a surface reaction pathway occurring above 473 K (high-barrier surface reaction pathway) for CO₂ hydrogenation to CH₃OH via surface intermediates of bri-HCOO*, CH₃O* and CH₃OH*, in which bri-HCOO* hydrogenation is the rate-limiting elementary surface reaction.

A Zn₂₀Zr₈₀ catalyst pretreated by 3 MPa CO₂ adsorption at 303 K and purged in Ar was found to exclusively exhibit CO₃* species whose hydrogenation reactivity was also studied in 3 MPa H₂. Weak gaseous CH₃OH signals emerge and then remain unchanged up to 473 K, then strong signals appear at 523 K and above (Fig. 3a, b). As demonstrated

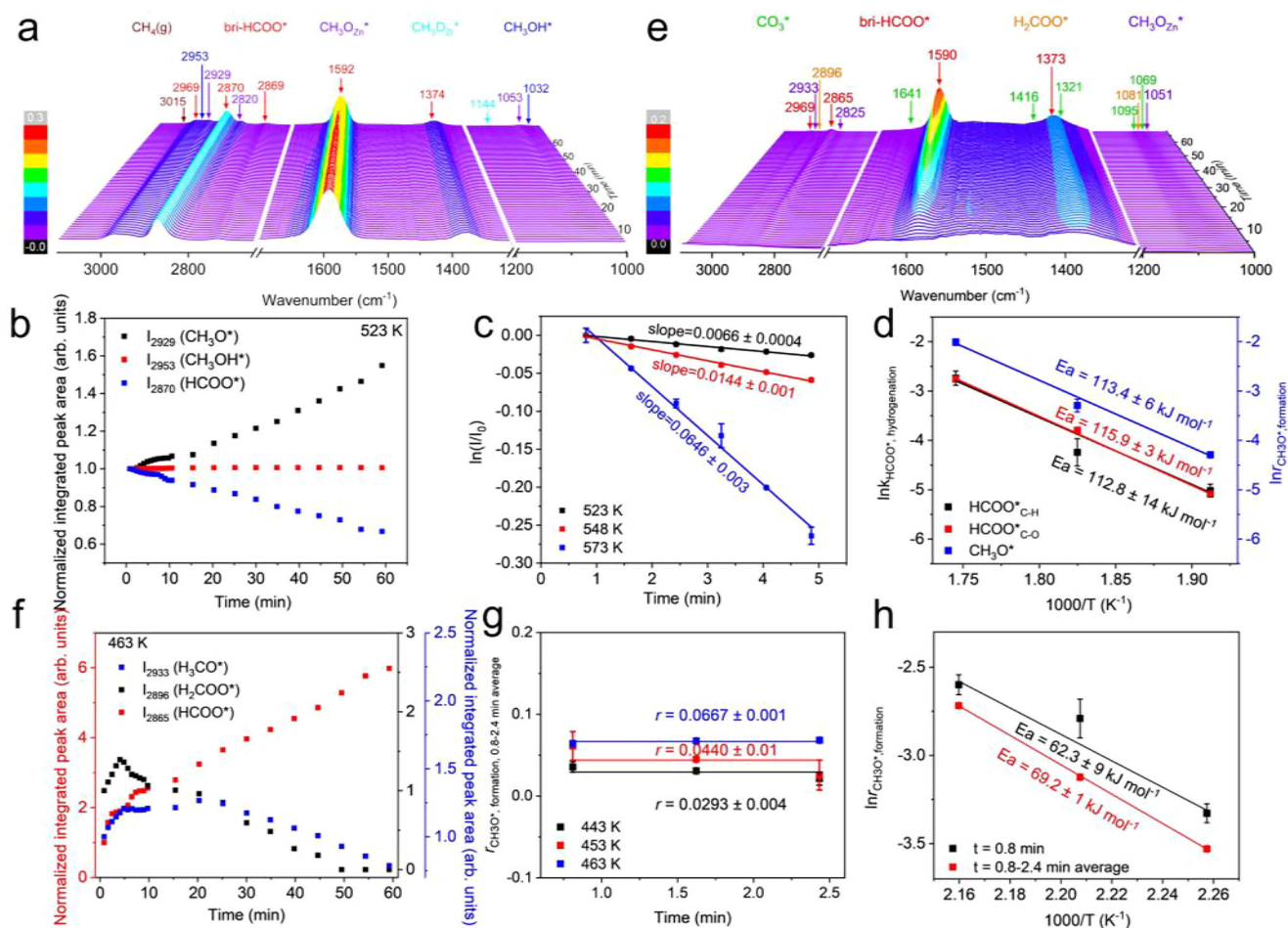


Fig. 4 | Elementary surface reaction kinetics at high and low temperatures in CO_2 hydrogenation reaction. **a** Temporal in situ DRIFTS spectra of Zn20Zr80 pretreated in 3 MPa $\text{CO}_2 + \text{H}_2$ at 573 K for 60 min and purged in Ar under 3 MPa H_2 at 523 K. **b** Corresponding normalized integrated peak areas of observed species in (a) as a function of time. **c** First-order reaction kinetic of bri-HCOO* hydrogenation reaction on Zn20Zr80 at different temperatures derived from panel (b), as well as Supplementary Figs. 24, 25. **d** Arrhenius plots of bridge formate hydrogenation and CH_3O^* formation reactions derived from Fig. 4c and Supplementary Figs. 27–28,

respectively. **e** Temporal in situ DRIFTS spectra of Zn20Zr80 pretreated in 3 MPa CO_2 at 303 K for 60 min and purged in Ar under 3 MPa H_2 at 463 K. **f** Corresponding normalized integrated peak areas of various species in (e) as a function of time. **g**, the average rate of CH_3O^* formation reaction on Zn20Zr80 at 0.8 to 2.4 min derived from panel (f). **h** Arrhenius plot of CH_3O^* formation reaction using the CH_3O^* formation rate at 0.8 min and the average rate of 0.8–2.4 min shown in Supplementary Fig. 38 and Fig. 4g. The error bars in the figure represent the standard errors (SE) of the fitted values.

in the corresponding in-situ DRIFTS spectra (Fig. 3e and Supplementary Fig. 14), CO_3^* weakens greatly at 423 K and disappears at 473 K; bri-HCOO* emerges at 303 K, grows slowly up to 423 K and rapidly up to 548 K, then weakens at 573 K; H_2COO^* obviously appears at 303 K, significantly grows at 373 K and then almost disappears at 423 K; both $\text{CH}_3\text{O}_{\text{Zn}}^*$ and CH_3OH^* vary similarly to H_2COO^* up to 423 K; additionally, $\text{CH}_3\text{O}_{\text{Zn}}^*$ re-emerges at 473 K, grows at 523 K and then weakens up to 573 K, while CH_3OH^* re-emerges at 523 K, does not change much at 548 K and almost disappears at 573 K; $\text{CH}_3\text{O}_{\text{Zr}}^*$ emerges at 473 K and grows slowly up to 573 K. Thus, in addition to the high-barrier surface reaction pathway, another surface reaction pathway occurring at 303 K and above (low-barrier surface reaction pathway) also exists for CO_2 hydrogenation to CH_3OH via surface intermediates of CO_3^* , bri-HCOO*, H_2COO^* , CH_3O^* and CH_3OH^* , in which CH_3O^* hydrogenation seems as the rate-limiting elementary surface reaction.

Elementary surface reaction kinetics of CO_2 hydrogenation reaction

Using sodium formate as the formate source, a calibration of the DRIFTS signal from formate species was done according to the previously-reported method⁵³, whose results (Supplementary Fig. 15) confirm a linear correlation between the formate concentration and

the integrated area of IR vibrational peak. The elementary surface reaction kinetics were studied by measuring temporal evolutions of various surface intermediates during the hydrogenation reaction at different temperatures using in situ DRIFTS. For the high-barrier reaction pathway, the Zn20Zr80 catalyst pretreated in 3 MPa $\text{CO}_2 + \text{H}_2$ at 573 K for 60 min, cooled down to 303 K and then purged in Ar was rapidly heated in 3 MPa H_2 to desired temperatures. As shown in Fig. 4a and Supplementary Figs. 16, 17, bri-HCOO* is the only species observed on the catalyst surface and it barely changes at temperatures up to at 473 K but then keeps weakening at higher temperatures, leading to the emergence of CH_3O^* and CH_3OH^* species. The C-H stretch vibration region of temporal in situ DRIFTS spectra at various temperatures were peak-fitted to give the corresponding evolutions of bri-HCOO*, CH_3O^* and CH_3OH^* species (Supplementary Figs. 18–23), represented respectively by the vibrational features at 2870, 2929 and 2953 cm^{-1} , against the reaction time (Fig. 4b and Supplementary Figs. 24, 25). The decreasing of bri-HCOO* species, which arises from its hydrogenation reaction, accelerates with the reaction temperature increasing. CH_3O^* keeps growing at 523 K, initially grows and then does not change much at 548 K, and initially grows and then weakens at 573 K. CH_3OH^* remains unchanged at 523 K, initially remains unchanged and then weakens at 548 and 573 K. Following the elementary surface reaction

kinetic model proposed in Supplementary Fig. 26 with the bri-HCOO^* hydrogenation reaction as the rate-limiting step, the initial growth of CH_3O^* species approximately equals to the CH_3O^* formation by the preceding bri-HCOO^* hydrogenation reactions. The bri-HCOO^* hydrogenation in 3 MPa H_2 were found to follow the first-order reaction kinetics (Fig. 4c), which is reasonable because the concentration of active H species, another reactant, could be considered constantly. The rate constants for bri-HCOO^* hydrogenation reaction and the rates for CH_3O^* formation reaction at 523, 548 and 573 K (Fig. 4c and Supplementary Fig. 27) were derived and used to plot the Arrhenius plots (Fig. 4d), which give activation energies of bri-HCOO^* hydrogenation reaction as $112.8 \pm 14 \text{ kJ mol}^{-1}$ and of CH_3O^* formation reaction as $113.4 \pm 6 \text{ kJ mol}^{-1}$. A similar analysis using the C-O vibrational feature of bri-HCOO^* at 1590 cm^{-1} (Supplementary Fig. 28) gives an activation energy of bri-HCOO^* hydrogenation reaction as $115.9 \pm 3 \text{ kJ mol}^{-1}$ (Fig. 4d). These results confirm that the bri-HCOO^* species hydrogenates to produce the CH_3O^* species. However, the activation energy of bri-HCOO^* hydrogenation reaction is much larger than the apparent activation energy of CO_2 hydrogenation to methanol reaction ($81.9 \pm 1 \text{ kJ mol}^{-1}$), therefore, the high-barrier surface reaction pathway is not likely to be responsible for the methanol synthesis from CO_2 hydrogenation catalyzed by Zn20Zr80 catalyst.

For the low-barrier reaction pathway, the Zn20Zr80 catalyst pretreated in 3 MPa CO_2 at 303 K for 60 min and then purged in Ar was rapidly heated in 3 MPa H_2 to 443, 453 and 463 K at which the high-temperature surface reaction pathway barely works. As shown in Fig. 4e and Supplementary Fig. 29, the temporal evolutions of CO_3^* , bri-HCOO^* , H_2COO^* and CH_3O^* species depend on the reaction temperatures. The C-H stretch vibration region of temporal in situ DRIFTS spectra at various temperatures were peak-fitted (Supplementary Figs. 30–35) to give the corresponding evolutions of bri-HCOO^* , H_2COO^* and CH_3O^* species, represented respectively by the vibrational features at 2865 , 2896 and 2933 cm^{-1} , against the reaction time (Fig. 4f and Supplementary Figs. 36, 37). bri-HCOO^* keeps increasing at all temperatures, whereas H_2COO^* initially increases but then decreases. These observations clearly demonstrate the existence of another type of bri-HCOO^* species (denoted as $\text{bri-HCOO}^*\text{-I}$) which exhibits vibrational features indistinguishable with those of the bri-HCOO^* species observed in the high-barrier reaction pathway (denoted as $\text{bri-HCOO}^*\text{-II}$) in our DRIFTS spectra but is capable of hydrogenating at low temperatures. CH_3O^* initially grows, then does not change at 443 and 453 K but weakened at 463 K. The CH_3OH^* species was not observed, probably due to the very low coverage. Following elementary surface reaction kinetic model proposed in Supplementary Fig. 39 with the CH_3O^* hydrogenation as the rate-limiting elementary surface reaction, the initial growth of CH_3O^* species approximately equals to the CH_3O^* formation by the preceding $\text{bri-HCOO}^*\text{-I}$ hydrogenation reactions. The rates for the CH_3O^* formation reaction at 443, 453 and 463 K were derived (Fig. 4g) and used to plot the Arrhenius plot (Fig. 4h) respectively, which gives activation energies varying between 62.3 ± 9 and $69.2 \pm 1 \text{ kJ mol}^{-1}$, slightly lower than the apparent activation energy of CO_2 hydrogenation to methanol ($81.9 \pm 1 \text{ kJ mol}^{-1}$). Thus, the low-barrier surface reaction pathway is predominantly responsible for the methanol synthesis from CO_2 hydrogenation catalyzed by the Zn20Zr80 catalyst.

It should be noted that the surface reactions of various surface intermediates on working Zn20Zr80 involved in the low-barrier reaction pathway proceed fast at typical reaction temperatures above 448 K, leading to their low coverages which are beyond the detection sensitivity of in situ DRIFTS during our studies. Nevertheless, the catalytic performance of Zn20Zr80 is predominantly contributed by the low-barrier reaction pathway. The surface intermediates on working Zn20Zr80 above 448 K observed by in situ

DRIFTS are those involved in the high-barrier reaction pathway, which, however, are spectators.

Theoretical simulations of CO_2 hydrogenation reaction

Comprehensive theoretical calculations were then carried out to examine the thermodynamics of the Zn-ZrO₂ surface configurations and possible CO_2 hydrogenation reaction pathways utilizing our recently-developed large-scale machine learning atomic simulation⁵⁴ and DFT computation. In accordance with the experimental observations of Zn_1 single atoms and Zn_n clusters on Zn20Zr80 oxide solid solution catalyst with a *t*-ZrO₂ phase, we adopted a slab model of the *t*-ZrO₂ (101) surface and replaced 1 to 3 Zr atoms on the surface with Zn, thus creating ZnZrO surfaces with Zn coverage percentages of 8.3%, 16.6%, and 25%, denoted as $\text{Zn}_1\text{-ZrO}_2$, $\text{Zn}_2\text{-ZrO}_2$, and $\text{Zn}_3\text{-ZrO}_2$, respectively. The surface O atoms were removed accordingly to maintain Zn with a valence of +2 and Zr with a valence of +4.

By exploring over 10,000 minima for each composition using SSW-NN, we obtained the global minimums of $\text{Zn}_1\text{-ZrO}_2$, $\text{Zn}_2\text{-ZrO}_2$ and $\text{Zn}_3\text{-ZrO}_2$ surfaces as shown in Fig. 5a–c, respectively. $\text{Zn}_3\text{-ZrO}_2$ was calculated more stable than $\text{Zn}_1\text{-ZrO}_2$ by 20.2 kJ mol^{-1} per Zn atom (Eq. (18)), suggesting a tendency for ZnO species to aggregate on ZrO₂ surfaces, in alignment with earlier simulation findings³¹. A more detailed examination reveals the presence of two distinct Zn sites: Zn-O-Zr (Zn_{Zr}) in all three $\text{Zn}_n\text{-ZrO}_2$ configurations and Zn-O-Zn (Zn_{Zn}) exclusively in $\text{Zn}_3\text{-ZrO}_2$. On the $\text{Zn}_1\text{-ZrO}_2$ surface (Fig. 5a), the most favorable location for introduced oxygen vacancies is in close proximity to the monodispersed Zn site. The coordination number of the neighboring surface Zn and two Zr atoms are 4 and 6, respectively, while the coordination number of other surface Zr atoms remains as 7, consistent with the pristine *t*-ZrO₂ (101) surface. On the $\text{Zn}_2\text{-ZrO}_2$ surface (Fig. 5b), the Zn atoms remain monodispersed, forming a linear arrangement along the [010] direction. Due to the increased density of monodispersed Zn atoms, Zn_{4c} and Zr_{6c} are neighbors. On the $\text{Zn}_3\text{-ZrO}_2$ surface (Fig. 5c), the Zn_{Zn} site could be regarded as aggregated ZnO species, with a Zn-Zn distance of $3.39 \text{ \AA}$, slightly longer than that in bulk ZnO ($3.21 \text{ \AA}$). The aggregation of ZnO on the surface results in additional oxygen vacancies, leading to all surface Zr atoms adopting a Zr_{6c} coordination.

The potential reaction pathways of CO_2 hydrogenation to methanol under 3 MPa $\text{CO}_2 + \text{H}_2$ ($\text{H}_2/\text{CO}_2 = 3$) at 573 K were explored at both the Zn_{Zr} and Zn_{Zn} sites through comprehensive DFT calculations of the Gibbs free energy (Eq. (17)). H_2 preferentially dissociates heterolytically at the Zn_{Zr} -O pair of the Zn-ZrO₂ surfaces to form $\text{H}^*\text{-Zn}_{\text{Zr}}$ and $\text{H}^*\text{-O}$ species with a low barrier of 23.2 kJ mol^{-1} . At the Zn_{Zr} site on the $\text{Zn}_1\text{-ZrO}_2$ surface (Fig. 5d and Supplementary Table 5), the reaction commences with CO_2 adsorption at the Zr_{6c} site and H_2 dissociation at the Zn_{Zr} -O pair. Subsequently, the $\text{H}^*\text{-Zn}_{\text{Zr}}$ species reacts with CO_2^* by surmounting a barrier of 34.7 kJ mol^{-1} (Fig. 5d, TS1), forming a $\text{bri-HCOO}_{\text{Zr,Zr}}^*$ intermediate with the O atom bonded to two Zr_{6c} cations. The $\text{bri-HCOO}_{\text{Zr,Zr}}^*$ species further hydrogenates with another $\text{H}^*\text{-Zn}_{\text{Zr}}$ species with a barrier of 30.9 kJ mol^{-1} to form the H_2COO^* intermediate (TS2). Then the H_2COO^* species hydrogenates with a weakly-adsorbed H_2 species into $\text{H}^*\text{-Zn}_{4c}$ and H_2COOH^* with a low barrier of 11.6 kJ mol^{-1} (TS3). The H_2COOH^* then reacted with $\text{H}^*\text{-Zn}_{\text{Zr}}$, forming OH^* and CH_3O^* with a barrier of 77.2 kJ mol^{-1} (TS4), which are then hydrogenated by overcoming a barrier of 41.5 kJ mol^{-1} (TS5) to H_2O and 86.8 kJ mol^{-1} (TS6) to methanol as the final product, respectively. On the $\text{Zn}_2\text{-ZrO}_2$ surface of a similar local Zn environment to the $\text{Zn}_1\text{-ZrO}_2$ surface, H_2 dissociation preferentially occurs at the Zn_{Zr} -O site, and interestingly, endows the identical Zr_{6c} sites A and B on a pristine surface (Fig. 5b) different local environment. Site A emerges on the opposite side of the oxygen vacancy, counter to the $\text{H}^*\text{-Zn}_{\text{Zr}}$ species, whereas the site B positions adjacently to the neighboring $\text{H}^*\text{-Zn}_{\text{Zr}}$ species. This leads to CO_2 adsorption with the presence of $\text{H}^*\text{-Zn}_{\text{Zr}}$

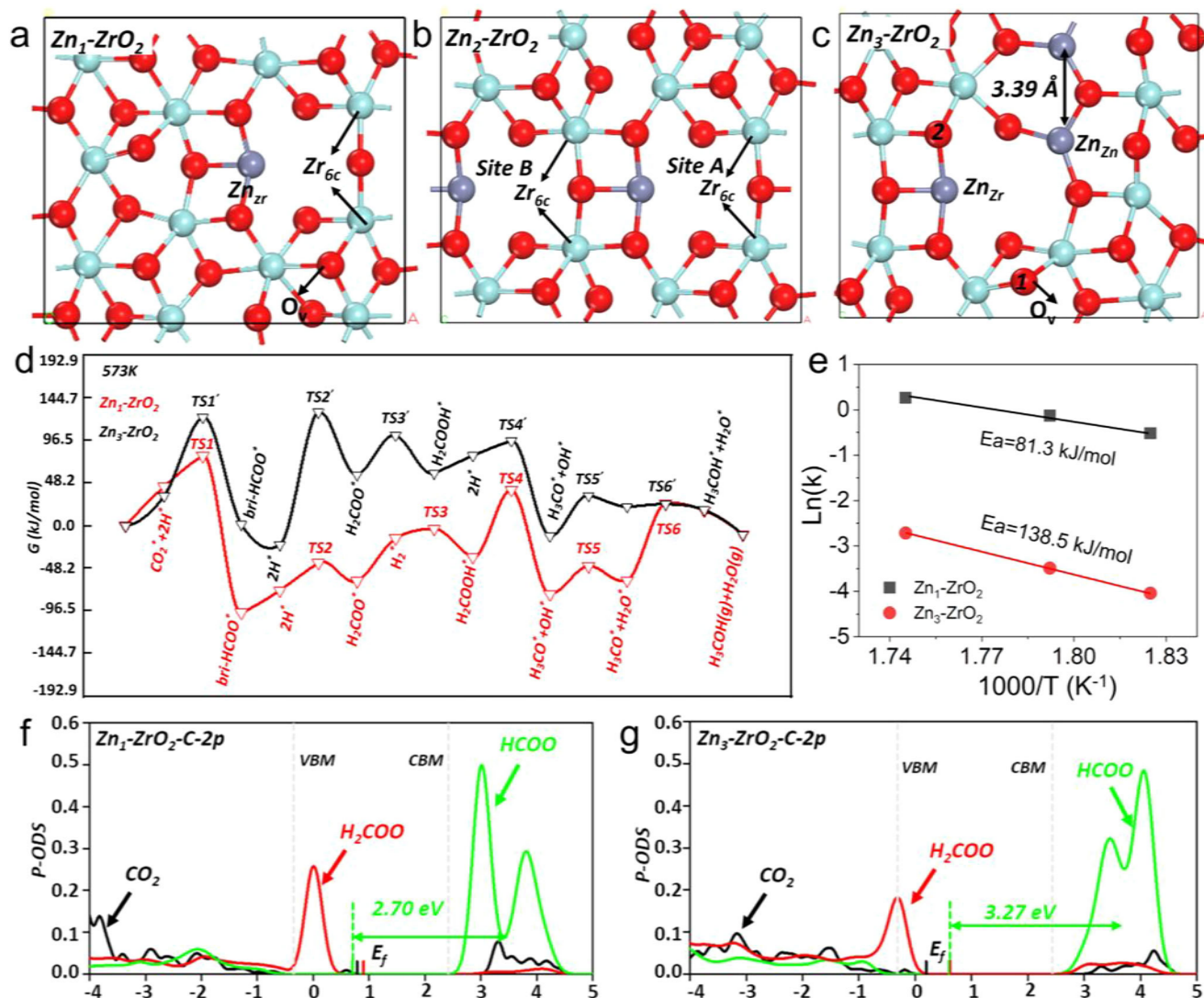


Fig. 5 | Theoretical calculations of CO₂ hydrogenation reaction. a–c Local snapshot of surface configurations of three global minimum structure with different surface Zn concentration, i.e., Zn₁-ZrO₂, Zn₂-ZrO₂ and Zn₃-ZrO₂. The O_v is the most stable site when the lattice O is removed. Key reaction sites are labeled out in each (d). The Gibbs free energy profile of CO₂ hydrogenation on two catalysts under reaction condition, i.e., Zn₁-ZrO₂ (red curve, 573 K, 3 MPa, H₂: CO₂ = 3) and Zn₃-ZrO₂ (black curve, 573 K, 3 MPa, H₂: CO₂ = 3). **e** Logarithm of methanol

production rates versus the reciprocal of temperatures on two catalysts. **f, g** The *pDOS* for the 2*p* orbitals associated with each carbon species (CO₂^{*}, HCOO^{*}, and H₂COO^{*}) on both Zn₁-ZrO₂ and Zn₃-ZrO₂ surfaces. All the DOSs are aligned via the Zr 3*d* states of the bulk *t*-ZrO₂. The *E_f* indicate the Fermi level; the vertical gray dotted lines indicate the valence band minimum (VBM) and conduction band minimum (CBM) of bulk ZrO₂, respectively.

species at site A with an adsorption energy of 107.1 kJ mol⁻¹ while at site B with an adsorption energy of 137 kJ mol⁻¹. The reaction between H^{*}-Zn_{Zr} and CO₂^{*} at site A or site B into the bri-HCOO_{Zr,Zr}^{*} exhibits an energy barrier of 85.9 and 139.9 kJ mol⁻¹, respectively. The energy profiles for the further hydrogenation reactions of bri-HCOO_{Zr,Zr}^{*} species at site A into CH₃O^{*} are similar to the Zn₁-ZrO₂ surface. On the Zn₃-ZrO₂ surface (Fig. 5d and Supplementary Table 6), H₂ dissociation also occurs at the Zn_{Zr}-O site, and CO₂ adsorbs bridgingly at the Zn_{Zr}-O-Zr_{6c} site. Then the CO₂^{*} species undergoes stepwise hydrogenation reactions into the bri-HCOO_{Zn,Zr}^{*}, H₂COO^{*}, H₂COOH^{*}, and H₃CO^{*} intermediates, in which the rate determining step and the overall barrier is the HCOO_{Zn,Zr}^{*} hydrogenation by H^{*}-Zn_{Zr} with a barrier as high as 146.7 kJ mol⁻¹.

Thus, although CO₂ hydrogenation to methanol follows the same reaction pathway on Zn₁-ZrO₂, Zn₂-ZrO₂ and Zn₃-ZrO₂, the rate-limiting step and associated barrier differ much. It is the hydrogenation of H₂COOH^{*} into CH₃O^{*} and OH^{*} with a barrier of 77.2 kJ mol⁻¹ and the hydrogenation of CH₃O^{*} into CH₃OH^{*} with a

barrier of 86.8 kJ mol⁻¹ on Zn₁-ZrO₂, the hydrogenation of CO₂^{*} into bri-HCOO_{Zr,Zr}^{*} with a barrier of 85.8 kJ mol⁻¹ and the hydrogenation of CH₃O^{*} into CH₃OH^{*} with a barrier of 96.5 kJ mol⁻¹ on Zn₂-ZrO₂, and the hydrogenation of bri-HCOO_{Zn,Zr}^{*} into H₂COOH^{*} with a barrier of 146.7 kJ mol⁻¹ on Zn₃-ZrO₂. Microkinetic simulations for CO₂ hydrogenation to methanol on Zn₁-ZrO₂ and Zn₃-ZrO₂ catalysts using the free energetics derived from DFT give an apparent activation energy of 81.3 kJ mol⁻¹ on Zn₁-ZrO₂ and of 138.5 kJ mol⁻¹ on Zn₃-ZrO₂ (Fig. 5e). These results suggest that the experimentally-observed low-barrier surface reaction pathway of CO₂ hydrogenation to methanol, in which the CH₃O^{*} formation reaction from the bri-HCOO^{*}-I species exhibits an activation energy of 69.6 kJ mol⁻¹ and the CH₃O^{*} hydrogenation reaction is the rate-limiting step, probably occurs at the Zn₁ single atom site of Zn₂₀Zr₈₀, while the experimentally-observed high-barrier surface reaction pathway, in which the bri-HCOO^{*}-II hydrogenation is the rate-limiting step with an activation energy of 112.8 kJ mol⁻¹, probably occurs at the Zn_n cluster sites. The experimentally-observed bri-HCOO^{*} species during CO₂

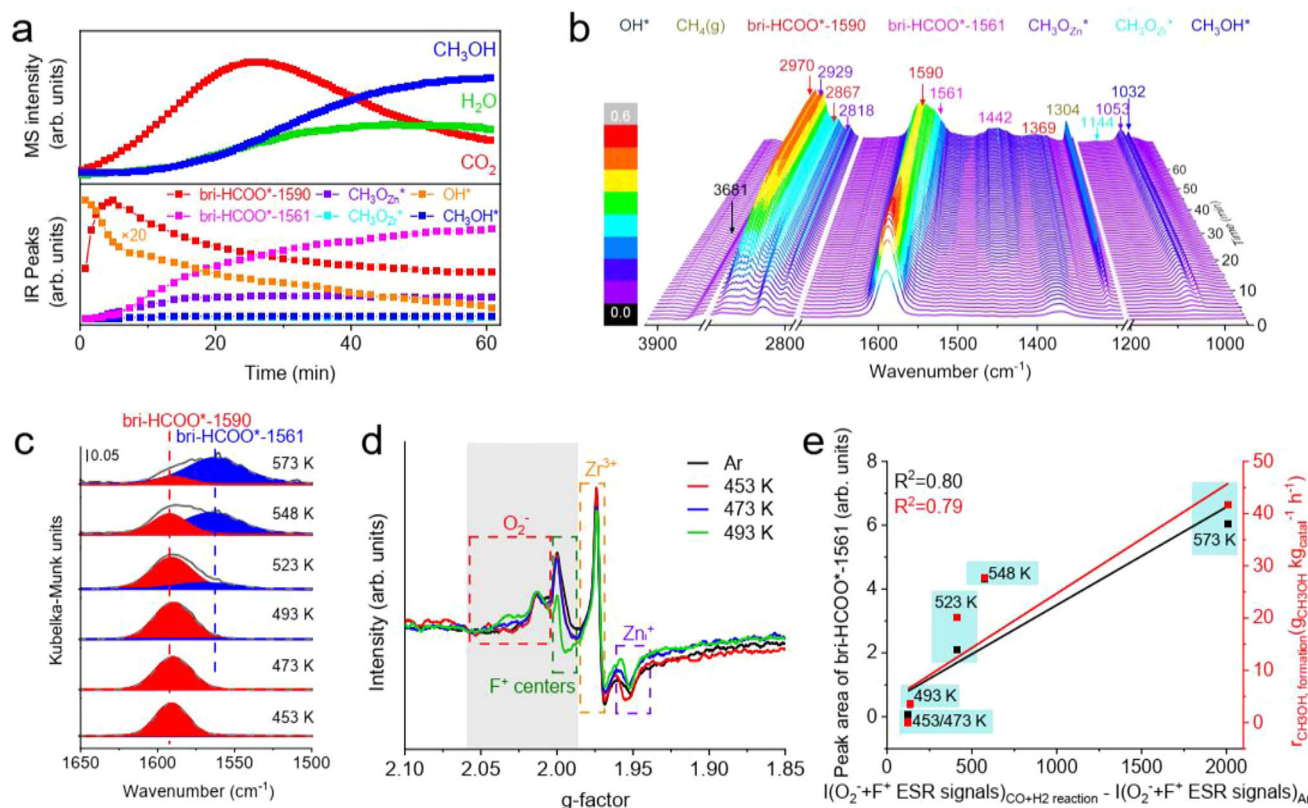


Fig. 6 | Operando characterizations and in situ restructuring in CO hydrogenation reaction. **a** CH_3OH MS signals and corresponding IR peaks of species in **(b)** as a function of time, and **b**, corresponding temporal in situ DRIFTS spectra of $\text{Zn}_{20}\text{Zr}_{80}$ under 3 MPa $\text{CO} + \text{H}_2$ ($\text{H}_2/\text{CO}=2$) atmosphere at 573 K. **c** Peak-fitted C-O vibrational region of steady-state in situ DRIFTS spectra of $\text{Zn}_{20}\text{Zr}_{80}$ under 3 MPa $\text{CO} + \text{H}_2$ at indicated temperatures. **d** ESR spectra measured at 140 K of $\text{Zn}_{20}\text{Zr}_{80}$

calced in Ar at 773 K and subsequently treated under 3 MPa $\text{CO} + \text{H}_2$ at indicated temperatures without exposure to air. **e**, intensity of vibrational feature of bri-HCOO^*-1561 species on $\text{Zn}_{20}\text{Zr}_{80}$ treated under 3 MPa $\text{CO} + \text{H}_2$ at different temperatures and CH_3OH formate rate of $\text{Zn}_{20}\text{Zr}_{80}$ -catalyzed CO hydrogenation reaction at different temperatures as a function of corresponding amount of created O_2^- and F^+ centers on $\text{Zn}_{20}\text{Zr}_{80}$ surface.

hydrogenation on $\text{Zn}_{20}\text{Zr}_{80}$ contains the reactive $\text{bri-HCOO}_{\text{Zr,Zr}}^*$ ($\text{bri-HCOO}^*\text{-I}$) species below 448 K while is dominantly the much less reactive $\text{bri-HCOO}_{\text{Zn,Zr}}^*$ ($\text{bri-HCOO}^*\text{-II}$) species above 448 K. Additional theoretical calculations using PBE-D3⁵⁵ to include the dispersion interaction (Supplementary Table 7) show that the dispersion interaction does not affect the computed reaction energies much.

To gain insights into the disparate catalytic behaviors of $\text{Zn}_1\text{-ZrO}_2$ and $\text{Zn}_3\text{-ZrO}_2$ surfaces, we performed the electronic structure analysis of various adsorbed species. Upon H_2 dissociation on both surfaces, the H^* of $\text{H}^*-\text{Zn}_{\text{Zr}}$ acquires a negative charge of around $-0.4|e|$, with the bonded Zn also gaining $-0.2|e|$, while the H^* of H^*-O acquires a positive charge of about $0.6|e|$. Thus, the reactivity of H^* species was similar on $\text{Zn}_1\text{-ZrO}_2$ and $\text{Zn}_3\text{-ZrO}_2$. The Projected Density of States (ρDOS) for the $2p$ orbitals associated with the CO_2^* , HCOO^* , and H_2COO^* on $\text{Zn}_1\text{-ZrO}_2$ and $\text{Zn}_3\text{-ZrO}_2$ were analyzed (Fig. 5f, g). Notably, an unoccupied state of the C_{2p} orbital was found to emerge above the Conduction Band Minimum (CBM) upon the hydrogenation of CO_2^* into HCOO^* and become subsequently occupied upon the hydrogenation of HCOO^* into H_2COO^* . This orbital exhibits a π -bond character on both $\text{Zn}_1\text{-ZrO}_2$ and $\text{Zn}_3\text{-ZrO}_2$ surfaces, and contains substantial contribution from the d orbitals of Zr atoms bonded to the HCOO^* intermediate but little contribution from the Zn atom bonded to the HCOO^* intermediate (Supplementary Fig. 40). As a result, the center of this unoccupied C_{2p} orbital band is positioned at 2.70 eV for the $\text{bri-HCOO}_{\text{Zr,Zr}}^*$ species on $\text{Zn}_1\text{-ZrO}_2$ and 3.27 eV for the $\text{bri-HCOO}_{\text{Zn,Zr}}^*$ species on $\text{Zn}_3\text{-ZrO}_2$ relative to the Fermi level. This leads to a greater difficulty of the $\text{bri-HCOO}_{\text{Zn,Zr}}^*$ species in acquiring electrons from negatively charged H^* of $\text{H}^*-\text{Zn}_{\text{Zr}}$ than the $\text{bri-HCOO}_{\text{Zr,Zr}}^*$ species, explaining the much higher

barrier of $\text{bri-HCOO}_{\text{Zn,Zr}}^*$ hydrogenation reaction than of $\text{bri-HCOO}_{\text{Zr,Zr}}^*$ hydrogenation reaction. We calculated the C-O vibrational mode of $\text{bri-HCOO}_{\text{Zn,Zr}}^*$ and $\text{bri-HCOO}_{\text{Zr,Zr}}^*$ species to be 1563 and 1556 cm^{-1} , respectively. Thus, it is difficult to experimentally distinguish the $\text{bri-HCOO}_{\text{Zn,Zr}}^*$ and $\text{bri-HCOO}_{\text{Zr,Zr}}^*$ species using vibrational spectroscopy. Based on our results, previous argument of $\text{bri-HCOO}_{\text{Zn,Zr}}^*$ species as the active formate intermediate in the CO_2 hydrogenation to methanol reaction catalyzed by the Zn-O-Zr active site³⁰ is wrong.

Reaction mechanism of CO hydrogenation reaction

ESR spectrum of $\text{Zn}_{20}\text{Zr}_{80}$ subjected to the CO hydrogenation reaction at 573 K without exposure to air exhibits substantially increased signals of O_2^- and F^+ centers but barely changed signals of Zr^{3+} and Zn^+ defects (Fig. 1d). Following a subsequent exposure to air, the signals of O_2^- and F^+ centers are quenched whereas those of Zr^{3+} and Zn^+ defects do not change (Supplementary Fig. 41), suggesting that the increased O_2^- and F^+ centers on $\text{Zn}_{20}\text{Zr}_{80}$ subjected to the CO hydrogenation reaction at 573 K should probably locate on the $\text{Zn}_{20}\text{Zr}_{80}$ surface. Thus, the $\text{Zn}_{20}\text{Zr}_{80}$ surface is partially reduced during the CO hydrogenation reaction at 573 K to form O_2^- and F^+ centers. CO hydrogenation reaction at 573 K catalyzed by $\text{Zn}_{20}\text{Zr}_{80}$ was operando characterized. An induction period of around 55 min was observed prior to the stable CH_3OH production (Fig. 6a), during which CO_2 and H_2O productions occurred, supporting the occurrence of partial surface reduction of $\text{Zn}_{20}\text{Zr}_{80}$. The vibrational features of bri-HCOO^* , CH_3O^* , CH_3OH^* and OH^* species (Supplementary Table 3) were observed to vary with the reaction time in the corresponding temporal

in situ DRIFTS (Fig. 6b), and their evolutions are shown in Fig. 6a. The OH* species initially decreases quickly and then slowly, and its consumption leads to the partial surface reduction of Zn₂₀Zr₈₀. The originally existing HCOO* species at 1590/1369 cm⁻¹ (denoted as bri-HCOO*-1590) initially increases quickly and then decreases slowly, while another new HCOO* species at 1561/1442 cm⁻¹ (denoted as bri-HCOO*-1561) emerges and keeps growing up to around 55 min. It is noteworthy that the formation of HCOO* species upon CO adsorption on Zn₂₀Zr₈₀ must involve one surface lattice oxygen site. The CH₃O* and CH₃OH* species emerge, grow and keep unchanged after around 15 min, demonstrating that both surface species reach the equilibrium states far before the HCOO* species and gaseous CH₃OH. Thus, the CH₃O* and CH₃OH* species should not be involved in the rate-limiting surface reaction step, while the HCOO* hydrogenation reaction is the rate-limiting surface reaction.

Steady-state in situ DRIFTS spectra of Zn₂₀Zr₈₀-catalyzed CO hydrogenation reaction at various temperatures (Fig. 6c and Supplementary Fig. 42a) show that, with the temperature increasing, the bri-HCOO*-1590 species grows and reaches the maximum at 493 K and then weakens, while the bri-HCOO*-1561 species emerges at 493 K and then keeps growing. The evolutions of CH₃O* and CH₃OH* species are similar to that of bri-HCOO*-1561 species, not to that of bri-HCOO*-1590 species. The ESR spectra without exposure to air (Fig. 6d and Supplementary Fig. 42b) show that the additional O₂⁻ and F⁺ centers features emerge on the used Zn₂₀Zr₈₀ catalysts at 493 K and grows with the reaction temperature. The amount of paramagnetic centers in a sample is proportional to the area under its absorption curve that can be obtained by an integral treatment of the ESR spectrum^{56,57}, following which the intensities of O₂⁻ and F⁺ centers signals on Zn₂₀Zr₈₀ calcined in Ar at 773 K ($I(\text{O}_2^- + \text{F}^+)_{\text{Ar}}$) and subsequently treated under 3 MPa CO + H₂ at different temperatures ($I(\text{O}_2^- + \text{F}^+)_{\text{CO+H}_2 \text{ reaction}}$) were acquired. The intensity of vibrational feature of bri-HCOO*-1561 species on Zn₂₀Zr₈₀ treated under 3 MPa CO + H₂ at different temperatures and the CH₃OH formate rate of Zn₂₀Zr₈₀-catalyzed CO hydrogenation reaction at different temperatures were found proportional to the corresponding amount of created O₂⁻ and F⁺ centers on Zn₂₀Zr₈₀ surface, represented by $I(\text{O}_2^- + \text{F}^+)_{\text{CO+H}_2 \text{ reaction}}/I(\text{O}_2^- + \text{F}^+)_{\text{Ar}}$ (Fig. 6e). These results suggest that the active site of Zn₂₀Zr₈₀ in catalyzing the methanol production from CO hydrogenation is related with the in situ formed surface oxygen vacancy with the bri-HCOO*-1561 species as the active formate species, rather than the bri-HCOO*-1590 species on the stoichiometric surface.

Individual activation of CO and H₂ on Zn₂₀Zr₈₀ at 573 K was further studied (Supplementary Fig. 43). An exposure to 3 MPa CO forms bri-HCOO*-1590 and gaseous CO₂, and subsequent purging in Ar and heating in 3 MPa H₂ result in the formation of CH₃O* at the expense of bri-HCOO*. An exposure to 3 MPa H₂ demonstrates the hydrogenation of originally existing CO₃* species to bri-HCOO*-1590 and CH₃O*, and subsequent purging in Ar and heating in 3 MPa CO leads to the growth of bri-HCOO*-1590. These observations, on one hand, demonstrate that the bri-HCOO*-1590 species is capable of hydrogenating to produce CH₃OH, on the other hand, suggest that the reduction of Zn₂₀Zr₈₀ surface responsible for the formation of bri-HCOO*-1561 species is more facilitated under the syngas condition than under the individual CO or H₂ condition.

The Zn₂₀Zr₈₀ catalysts exclusively with the bri-HCOO*-1590 species or with the coexisting bri-HCOO*-1590 and bri-HCOO*-1561 species were prepared by the treatments in 3 MPa CO + H₂ (H₂:CO = 2) at 473 or 573 K for 60 min, respectively, following by cooling down to 303 K and purging in Ar (Supplementary Fig. 44a, b). Both formate species undergo the hydrogenation reaction in 3 MPa H₂ at 573 K. The hydrogenation reaction on the Zn₂₀Zr₈₀ catalyst exclusively with the bri-HCOO*-1590 species produces CO₂ but few CH₃OH, whereas the hydrogenation reaction on the Zn₂₀Zr₈₀ catalyst with the

coexisting bri-HCOO*-1590 and bri-HCOO*-1561 species produces significantly more CH₃OH (Supplementary Fig. 44c, d). This further supports that the bri-HCOO*-1561 species is responsible for the CH₃OH production from CO hydrogenation over Zn₂₀Zr₈₀.

Elementary surface reaction kinetics of CO hydrogenation reaction

The hydrogenation reaction on the Zn₂₀Zr₈₀ catalyst with the coexisting bri-HCOO*-1590 and bri-HCOO*-1561 specie in 3 MPa H₂ at various temperatures was characterized using temporal in situ DRIFTS spectra (Fig. 7a and Supplementary Fig. 45). The C-O stretch vibration peaks were peak-fitted (Supplementary Figs. 46–51) to give the evolutions of bri-HCOO*-1590 and bri-HCOO*-1561 species as a function of reaction time, respectively (Fig. 7b and Supplementary Fig. 53). Following the elementary surface reaction kinetic model proposed in Supplementary Fig. 52, the hydrogenation reaction of bri-HCOO*-1590 and bri-HCOO*-1561 was found to follow the first-order reaction kinetics (Fig. 7c, d), which is reasonable because the concentration of active H species during the hydrogenation reaction could be considered constant. Then the rate constants at various temperatures were calculated and consequently the Arrhenius plots (Fig. 7e) were plotted. The activation energy of bri-HCOO*-1590 and bri-HCOO*-1561 hydrogenation reactions was calculated as 121.6 ± 19 and 67.6 ± 9 kJ mol⁻¹, respectively. The rate for bri-HCOO*-1590 and bri-HCOO*-1561 hydrogenation reactions were also derived (Supplementary Fig. 54) and gives the activation energy of as 107.8 ± 17 and 68.9 ± 1 kJ mol⁻¹ (Supplementary Fig. 55), respectively. Comparing the apparent activation energy of CO hydrogenation to methanol reaction catalyzed by Zn₂₀Zr₈₀ (54.5 kJ mol⁻¹), the elementary surface reaction kinetics clearly demonstrate that the bri-HCOO*-1561 species related with the surface defective sites in situ created on Zn₂₀Zr₈₀ is the active formate species for the CH₃OH production, whereas the bri-HCOO*-1590 species formed on the stoichiometric Zn₂₀Zr₈₀ surface is a spectator.

The bri-HCOO*-1590 species barely hydrogenates at 453 K (Supplementary Fig. 56), thus its increase on Zn₂₀Zr₈₀ in 3 MPa CO + H₂ at various temperatures below 453 K exclusively arises from its formation by CO hydrogenation, whose temporal in situ DRIFTS spectra were measured (Supplementary Fig. 57). Following the elementary surface reaction kinetic model proposed in Supplementary Fig. 58, the formation rates of bri-HCOO*-1590 species at different temperatures were acquired and used to plot the Arrhenius plot (Supplementary Fig. 59), whose results give the activation energy of bri-HCOO*-1590 formation reaction by CO hydrogenation as 34.3 ± 6 kJ mol⁻¹. A treatment of Zn₂₀Zr₈₀ in 3 MPa CO was also found capable of forming the bri-HCOO*-1590 species at the expense of originally-existing OH* species. Using the temporal in situ DRIFTS spectra (Supplementary Fig. 60) to measure the elementary surface reaction kinetics at various temperatures below 453 K (Supplementary Fig. 61), the activation energy of bri-HCOO*-1590 formation reaction and OH* consumption was calculated as 36.3 ± 3 and 37.5 ± 1 kJ mol⁻¹ (Supplementary Fig. 62). These results demonstrate that the OH* is capable of hydrogenating CO to produce the bri-HCOO*-1590 species on Zn₂₀Zr₈₀.

Theoretical simulations of CO hydrogenation reaction

We firstly investigated the formation energy of an O vacancy (V_o) under the reaction conditions on both Zn₁-ZrO₂ and Zn₃-ZrO₂ surfaces, corresponding to a surface concentration of 0.0832 ML corresponding to stoichiometry *t*-ZrO₂ (110) surface. The most stable V_o locates at the Zr-O-Zr site on Zn₁-ZrO₂ and Zn₃-ZrO₂ (Fig. 5a, c), whose formation energy are both 1.39 eV. However, by enlarging the surface to a 2×2 supercell that gives a V_o concentration of 0.0208 ML, the formation energy of V_o is then reduced to 0.91 eV on Zn₁-ZrO₂ and 0.30 eV on Zn₃-ZrO₂. These results suggest that the surface with aggregated Zn species is more likely to be reduced, which is consistent with the recent

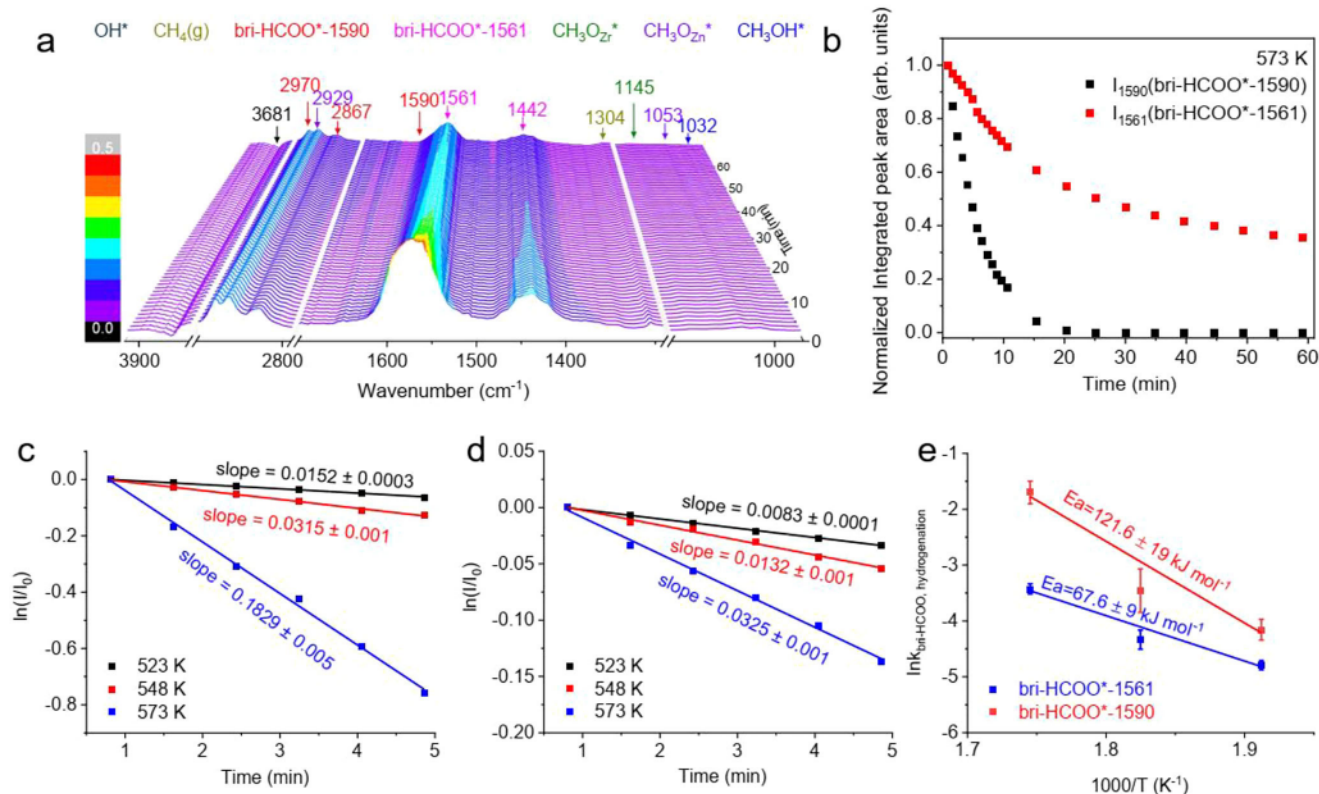


Fig. 7 | Elementary surface reaction kinetics at high temperatures in CO hydrogenation reaction. **a** Temporal in situ DRIFTS spectra of Zn20Zr80 (pretreated in 3 MPa CO + H₂, H₂:CO = 2, at 573 K for 60 min and purged in Ar) under 3 MPa H₂ at 523 K. **b** The corresponding normalized integrated peak area of observed species in Fig. 7a as a function of time on Zn20Zr80 catalyst at 573 K.

c First-order reaction kinetic of bri-HCOO*-1590 and **d**, bri-HCOO*-1561 hydrogenation reaction on Zn20Zr80 at different temperatures derived from panel (b), as well as Supplementary Fig. S3. **e** Arrhenius plots of bri-HCOO*-1590 and bri-HCOO*-1561 hydrogenation reactions derived from (c, d). The error bars in the figure represent the standard errors (SE) of the fitted values.

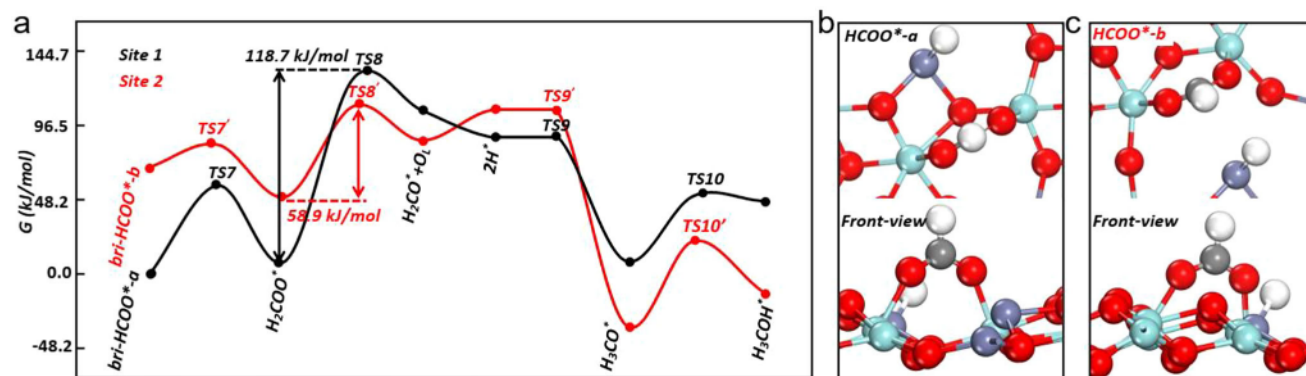


Fig. 8 | Theoretical calculations of CO hydrogenation reaction. **a** The energy profile of bri-HCOO* hydrogenation to CH₃OH at the most stable site (denoted as bri-HCOO*-a) and second most stable site (denoted as bri-HCOO*-b) of Zn₃-ZrO₂ surface. **b**, **c** The snapshots of bri-HCOO*-a and bri-HCOO*-b.

finding that the surface oxygen of ZnO will facilitate CO into CO₂⁵⁸.

Then the possible pathways of CO hydrogenation into methanol were explored by firstly considering the possible pathways for surface hydroxyl reacting with adsorbed CO molecules (Supplementary Fig. 63). On both Zn₁-ZrO₂ and Zn₃-ZrO₂ surfaces, the most stable adsorption site for CO is at the Zr top site, where the CO molecule inserts into a Zr-O bond with the C atom to form an OCO* species, giving the adsorption energy of -24.1 and -31.8 kJ mol⁻¹, respectively. The optimal pathway of HCOO* formation is through a diffusion of hydrogen from hydroxyl group to the nearby Zn site, followed by reacting with OCO*, to form a bri-HCOO*_{Zr,Zr} intermediate. The calculated barriers of H diffusion and bri-HCOO* formation is 60.8 and

183.3 kJ mol⁻¹ on Zn₁-ZrO₂, respectively, and 46.3 and 30.9 kJ mol⁻¹ on Zn₃-ZrO₂, respectively, while, as shown above, the experimentally-measured activation energies of hydroxyl consumption and bri-HCOO* formation during CO adsorption on the Zn20Zr80 catalyst respectively are 37.5 and 36.3 kJ mol⁻¹. Thus, the DFT calculation results indicate that the experimentally-observed HCOO* formation by CO reaction with OH groups should occur at the Zn_n site of Zn20Zr80 surface rather than at the Zn₁ site. The resulting bri-HCOO*_{Zr,Zr} species on Zn₃-ZrO₂ (denoted as bri-HCOO*-a, Fig. 8a, b) hydrogenates with H*-Zn_{Zr} species via a barrier of 56.9 kJ mol⁻¹ to form H₂COO*-a, and the C-O bond of H₂COO*-a subsequently breaks to form H₂CO*-a and a lattice oxygen via a barrier of 118.7 kJ mol⁻¹ (Fig. 8a and Supplementary Fig. 64), then H₂CO*-a hydrogenates with H*-Zn_{Zr}

barrierlessly, forming $\text{CH}_3\text{O}^*\text{-a}$, which further hydrogenates to produce CH_3OH via a barrier of 42.5 kJ mol^{-1} . The overall barrier of $\text{bri-HCOO}^*\text{-a}$ hydrogenation to CH_3OH is 128 kJ mol^{-1} , similar to the experimentally-determined barrier of $\text{bri-HCOO}^*\text{-1590}$ hydrogenation on unreduced $\text{Zr}_{20}\text{Zr}_{80}$ surface under CO hydrogenation reaction condition.

We note that the most stable adsorption site of CO on $\text{Zn}_3\text{-ZrO}_2$ surface is also the most reducible site and hence the most stable formate intermediate ($\text{bri-HCOO}^*\text{-a}$) may not present under the reductive condition of CO hydrogenation reaction. Consequently, we investigated the hydrogenation activity of the second most stable $\text{bri-HCOO}^*\text{-Zr}_{\text{Zr}}$ species (denoted as $\text{bri-HCOO}^*\text{-b}$, Fig. 8c), 69.5 kJ mol^{-1} less stable than the $\text{bri-HCOO}^*\text{-a}$ species. The formation energy of V_O ($134.1 \text{ kJ mol}^{-1}$) is relatively high in the adopted model, but we found that the Zn_{Zr} site gains 1.1e| with the presence of $\text{bri-HCOO}^*\text{-a}$, which, compared to the Zn_{Zr} on clean surface, mimics a reduced $\text{Zn}_3\text{-ZrO}_2$ surface for $\text{bri-HCOO}^*\text{-b}$ further hydrogenation reactions. As shown in Fig. 8a, on the $\text{Zn}_3\text{-ZrO}_2$ surface with the presence of $\text{bri-HCOO}^*\text{-a}$, $\text{bri-HCOO}^*\text{-b}$ hydrogenates to produce methanol following the same reaction pathway as $\text{bri-HCOO}^*\text{-a}$, however, the barrier of C-O bond breaking in $\text{H}_2\text{COO}^*\text{-b}$ to form $\text{H}_2\text{CO}^*\text{-b}$ is only 58.9 kJ mol^{-1} , being much lower than that in $\text{H}_2\text{COO}^*\text{-a}$. Accordingly, the overall barrier of $\text{bri-HCOO}^*\text{-b}$ hydrogenation into methanol is also 58.9 kJ mol^{-1} , similar to the experimentally-determined barrier of $\text{bri-HCOO}^*\text{-1561}$ hydrogenation on in situ partially-reduced $\text{Zr}_{20}\text{Zr}_{80}$ surface under CO hydrogenation reaction condition.

The $\text{bri-HCOO}^*\text{-a}$ species and $\text{bri-HCOO}^*\text{-b}$ species exhibit the calculated C-O vibrational mode at 1562 and 1549 cm^{-1} , respectively, and consequently correspond to the experimentally-observed $\text{bri-HCOO}^*\text{-1590}$ spectator and $\text{bri-HCOO}^*\text{-1561}$ active species on $\text{Zr}_{20}\text{Zn}_{80}$ for CO hydrogenation to methanol reaction. Thus, these DFT calculation results are consistent with the experimental results, supporting that the Zn_n cluster bonded to an in situ formed $\text{-Zr-V}_\text{O}\text{-Zr-}$ structure ($\text{-Zn-O-Zn(-O-Zr-V}_\text{O}\text{-Zr)-O-Zr-O-Zr-}$) ($\text{Zn}_{\text{n,OV}}$) on $\text{Zn}_{20}\text{Zr}_{80}$ is the active site for catalyzing the CO hydrogenation to methanol reaction and the associated $\text{bri-HCOO}^*\text{-1561}$ species is the active formate species whereas the unreduced Zn_n cluster and Zn_1 site are not and the associated $\text{bri-HCOO}^*\text{-1590}$ are spectator. As shown above, the bri-HCOO^* species at the unreduced $\text{Zn}_3\text{-ZrO}_2$ and $\text{Zn}_1\text{-ZrO}_2$ sites display similar C-O vibrational modes.

It is noteworthy that the CO hydrogenation to methanol reaction pathway at the in situ formed $\text{Zn}_{\text{n,OV}}$ active site does not exist in the $\text{Zn}_{20}\text{Zr}_{80}$ -catalyzed CO_2 hydrogenation reaction due to the absence of $\text{Zn}_{\text{n,OV}}$ active site. This exemplifies the high sensitivity of surface structure of working catalyst to the reaction atmosphere and consequently the active site structure. It is also noteworthy that the bri-HCOO^* species observed on $\text{Zn}_{20}\text{Zr}_{80}$ during CO_2 or CO hydrogenation to methanol reaction are with different structures and consequently formation ability and surface reactivity although they exhibit similar vibrational features. On one hand, both O atoms of bri-HCOO^* species come from CO_2 for CO_2 adsorption while one O atom comes from CO and the other comes from $\text{Zn}_{20}\text{Zr}_{80}$ surface for CO adsorption; on the other hand, the local structure of surface site on $\text{Zn}_{20}\text{Zr}_{80}$ binding the bri-HCOO^* species is different. The bri-HCOO^* species is commonly observed as a surface intermediate in the vibrational spectra characterizing oxide catalysts for CO_2 or CO hydrogenation to methanol reaction, but its structures and surface reactivity need to be carefully identified in order to determine the role.

$\text{Zn}_1/\text{m-ZrO}_2$ single atom catalyst

The above combined experimental and theoretical calculation results demonstrate that the Zn_1 -single atom ($\text{-Zr-O-Zn-O-Zr-O-Zr-}$) on stoichiometric $\text{Zn}_{20}\text{Zr}_{80}$ and the Zn_n cluster bonded to an in situ formed $\text{-Zr-V}_\text{O}\text{-Zr-}$ structure ($\text{-Zn-O-Zn(-O-Zr-V}_\text{O}\text{-Zr)-O-Zr-O-Zr-}$) on $\text{Zn}_{20}\text{Zr}_{80}$ are the active site for catalyzing the CO_2 and CO hydrogenation to

methanol reactions, respectively. Inspired by these findings, a $\text{Zn}_1/\text{m-ZrO}_2$ catalyst exclusively with the Zn_1 single atoms was prepared using a strong electrostatic adsorption method⁵⁹. The Zn loading was measured as 1.48 at. % and the BET specific surface area was measured as $22.6 \text{ m}^2/\text{g}$. XRD pattern of $\text{Zn}_1/\text{m-ZrO}_2$ (Supplementary Fig. 65) matches the standard pattern of pure monoclinic ZrO_2 (JCPDS no.88-2390). The Zn K-edge FT-EXAFS spectrum (Fig. 9a, Supplementary Fig. 66 and Supplementary Table 8) reveals that $\text{Zn}_1/\text{m-ZrO}_2$ barely has the Zn-Zn coordination shell, and the HAADF-STEM and EDS mapping images and corresponding single-pixel line profiles (Fig. 9b, c and Supplementary Fig. 67) demonstrate the Zn atoms are singly dispersed on ZrO_2 . The XPS spectrum shows that valence state of Zn in $\text{Zn}_1/\text{m-ZrO}_2$ is Zn^{2+} (Supplementary Fig. 4), while the Zn K-edge XANES spectrum (Fig. 9d) demonstrates that the Zn K-pre-edge feature of $\text{Zn}_1/\text{m-ZrO}_2$ lies between those of Zn foil and ZnO powder.

Catalytic performance of $\text{Zn}_1/\text{m-ZrO}_2$ in the CO_2 and CO hydrogenation reactions was evaluated (Fig. 9e). The $\text{Zn}_1/\text{m-ZrO}_2$ catalyst is greatly more active than the $\text{Zn}_{20}\text{Zr}_{80}$ catalyst in catalyzing the CO_2 hydrogenation to methanol reaction but much less active in catalyzing the CO hydrogenation to methanol reaction, while the CH_3OH selectivity of CO_2 or CO hydrogenation reaction catalyzed by $\text{Zn}_1/\text{m-ZrO}_2$ is similar to that by $\text{Zn}_{20}\text{Zr}_{80}$ at the same temperatures. The Arrhenius plots using CH_3OH formation rate catalyzed by $\text{Zn}_1/\text{m-ZrO}_2$ (Supplementary Fig. 68) give an E_a of $82.4 \pm 3 \text{ kJ mol}^{-1}$ in the CO_2 hydrogenation reaction, similar to that catalyzed by $\text{Zn}_{20}\text{Zr}_{80}$ ($78.9 \pm 4 \text{ kJ mol}^{-1}$), and an E_a of $84.2 \pm 17 \text{ kJ mol}^{-1}$ in the CO hydrogenation reaction, much larger than that catalyzed by $\text{Zn}_{20}\text{Zr}_{80}$ ($54.5 \pm 6 \text{ kJ mol}^{-1}$). The used $\text{Zn}_1/\text{m-ZrO}_2$ catalyst exhibits a similar structure to the fresh one. These observations further confirm that the active site on the $\text{Zn}_{20}\text{Zr}_{80}$ surface for catalyzing the CO_2 hydrogenation to methanol is the Zn_1 atom but not for catalyzing the CO hydrogenation to methanol. However, a $\text{Zn}_1/\text{t-ZrO}_2$ catalyst exhibits much poorer catalytic performance in the CO_2 hydrogenation to methanol reaction (Supplementary Fig. 69), indicating that the ZrO_2 crystal phase strongly affect the catalytic performance of Zn_1/ZrO_2 catalysts in the CO_2 hydrogenation reaction.

The H_2 TPR profiles (Fig. 9f) show that $\text{Zn}_1/\text{m-ZrO}_2$ is less reducible than $\text{Zn}_{20}\text{Zr}_{80}$, consistent with the DFT finding that the oxygen vacancy at the Zn_1 site is more difficult to be created than at the Zn_n site. This implies that the poor catalytic performance of $\text{Zn}_1/\text{m-ZrO}_2$ in the CO hydrogenation reaction probably results from its difficulty to be reduced and consequently the few surface oxygen vacancies. Reaction mechanisms of $\text{Zn}_1/\text{m-ZrO}_2$ catalyzed CO_2 or CO hydrogenation reaction were studied using in situ DRIFTS (Supplementary Figs. 70–72). In the CO_2 hydrogenation reaction, the low-barrier surface reaction pathway involving the key H_2COO^* intermediate was identified at 303 K and above but the high-barrier surface reaction pathway was not, consistent with the exclusive presence of Zn_1 single atom sites on $\text{Zn}_1/\text{m-ZrO}_2$. In the CO hydrogenation reaction, the active $\text{bri-HCOO}^*\text{-1561}$ species was not observed, consistent with the few surface oxygen vacancies on $\text{Zn}_1/\text{m-ZrO}_2$.

Discussion

In summary, using experimentally-measured elementary surface reaction kinetics to bridge the comprehensive experimental and theoretical calculation results, we unambiguously identify the active sites and associated reaction mechanisms of ZnO-ZrO_2 catalysts in the CO_2 or CO hydrogenation to methanol reaction. The Zn_1 single atom exclusively with the Zn-O-Zr (-Zr-O-Zn-O-Zr-) local structure and the Zn_n clusters with both the Zn-O-Zr and Zn-O-Zn ($\text{-Zn-O-Zn-O-Zr-O-Zr-}$) local structures coexist on the $\text{Zn}_{20}\text{Zr}_{80}$ surface. In the CO_2 hydrogenation to methanol reaction, the Zn_1 single atom is the active site with the CH_3O^* hydrogenation as the rate-limiting step. Interestingly, the experimentally-observed bri-HCOO^* intermediate on the working $\text{Zn}_{20}\text{Zr}_{80}$ catalyst at the typical reaction temperatures is predominantly the $\text{bri-HCOO}^*\text{-Zn}_{\text{Zr}}$ spectator at the Zn_n sites, but not the

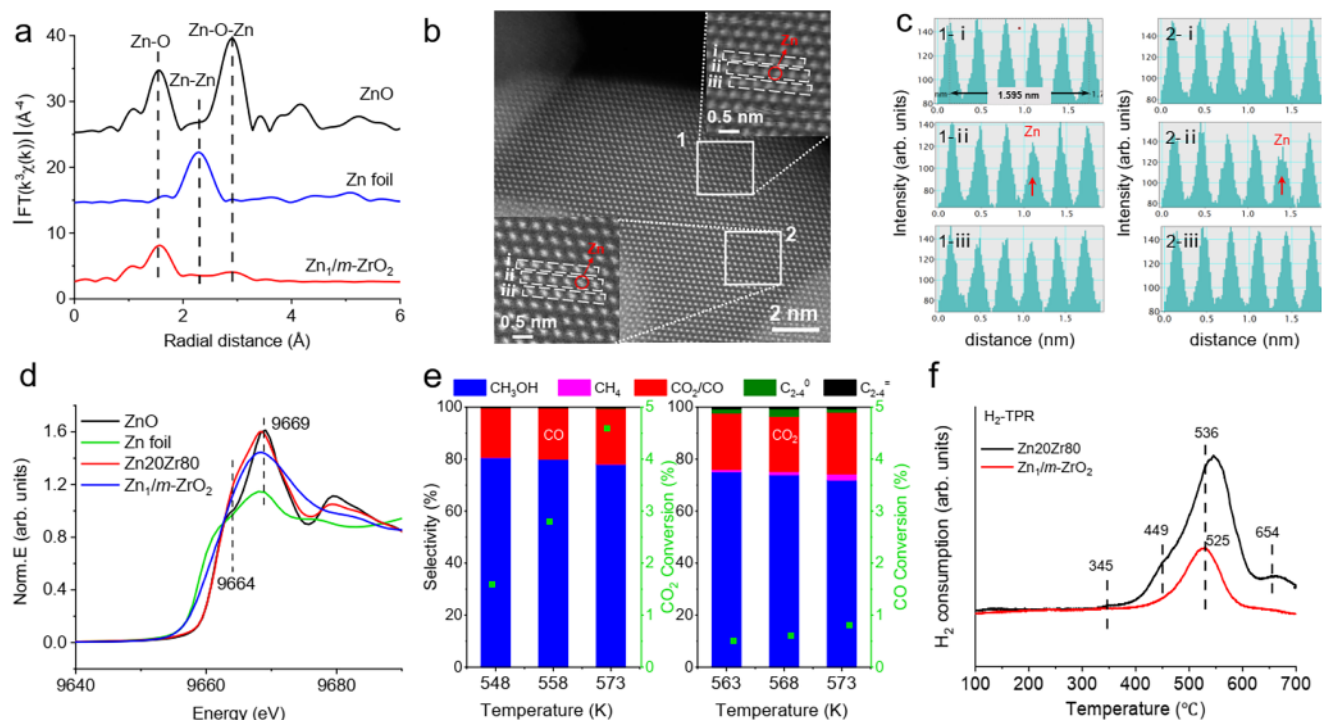


Fig. 9 | Structural characterizations and catalytic performance of $\text{Zn}_1/m\text{-ZrO}_2$ catalyst. **a** Fourier transforms of the k^3 -weighted Zn K-edge EXAFS spectra of $\text{Zn}_1/m\text{-ZrO}_2$ catalyst and referring ZnO and Zn foil. **b** Atomically resolved STEM spectrum imaging the edge of the $\text{Zn}_1/m\text{-ZrO}_2$ nanoparticle. **c** The corresponding single-pixel line profiles across the atom rows marked by rectangles in panel 1-i, 1-ii, 1-iii and 2-i, 2-ii, 2-iii. The dark dots and low-intensity single-pixel line profiles correspond to Zn-containing atomic columns, marked with red circles and arrows.

d Zn K-edge XANES spectra of $\text{Zn}_1/m\text{-ZrO}_2$ and Zn_2O catalyst and referring ZnO and Zn foil. **e** Catalytic performance of $\text{Zn}_1/m\text{-ZrO}_2$ catalyst in CO_2 (catalyst mass: 300 mg; flow rate: 30 mL min^{-1} ; $P = 3.0$ MPa; H_2 : $\text{CO}_2 = 3$) and CO hydrogenation (catalyst mass: 500 mg; flow rate: 16 mL min^{-1} ; $P = 3.0$ MPa; H_2 : $\text{CO} = 2$). **f** H_2 Temperature Programmed Reduction (H_2 -TPR) results of Zn_2O and $\text{Zn}_1/m\text{-ZrO}_2$ catalyst.

active $\text{bri-HCOO}_{\text{Zr,Zr}}^*$ species at the Zn_1 sites whose coverage is too low to be observed. In the CO hydrogenation to methanol reaction, the surface oxygen vacancies are created at the Zn_n sites but seldom at the Zn_1 sites, and the $\text{Zn}_{n,\text{OV}}$ site bonded to the in situ formed $-\text{Zr}-\text{O}-\text{Zr}-$ structure ($-\text{Zn}-\text{O}-\text{Zn}(-\text{O}-\text{Zr}-\text{V}_\text{O}-\text{Zr})-\text{O}-\text{Zr}-\text{O}-\text{Zr}-$) is the active site with the $\text{bri-HCOO}_{\text{Zr,Zr}}^*-\text{Zn}_{n,\text{OV}}$ hydrogenation as the rate-limiting step. The experimentally-observed $\text{bri-HCOO}_{\text{Zr,Zr}}^*-\text{Zn}_n$ and $\text{bri-HCOO}_{\text{Zr,Zr}}^*-\text{Zn}_1$ species on the working Zn_2O catalyst is the spectator. The distinctly different active sites of Zn_2O catalyst in the CO_2 and CO hydrogenation to methanol reactions illustrate high sensitivity of the active site structure of a catalyst to the reaction atmosphere. Nevertheless, the elementary surface reaction kinetics of working catalysts as a reliable correlation between experimental and theoretical calculation results provides a reliable methodology to unambiguously identify the active site of a catalyst in complex reactions.

Methods

Catalyst preparation

A co-precipitation method was adopted to synthesize the Zn_2O composite oxide. Typically, the measured amounts of $\text{Zr}(\text{NO}_3)_4 \cdot 5\text{H}_2\text{O}$ (National Pharmaceutical Chemical Reagent Co., Ltd., $\geq 98.0\%$) and $\text{Zn}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$ (Chinese Pharmaceutical Chemical Reagent Co., Ltd., $\geq 99.0\%$) were dissolved in 100 mL deionized water (resistivity $\geq 18.25 \text{ M}\Omega \cdot \text{cm}$), to which another 100 mL aqueous solution containing 3.06 g $(\text{NH}_4)_2\text{CO}_3$ (Chinese Pharmaceutical Chemical Reagent Co., Ltd., $\geq 99.999\%$) was added dropwise at a rate of 3 mL/min under vigorous stirring at 70°C . Then the mixture was stirred for an additional hour and aged for 3 h 70°C , followed by cooling to room temperature. The resulting precipitate was filtered, washed with deionized water for three times, then dried at 80°C for 16 h and finally calcined at 500°C for 3 h in air to acquire the Zn_2O sample. The same precipitation

method without an addition of $\text{Zn}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$ was also used to synthesize the $m\text{-ZrO}_2$ sample.

A Strong Electrostatic Adsorption (SEA) method⁵⁹ was adopted to synthesize the $\text{Zn}_1/m\text{-ZrO}_2$ catalyst using the $m\text{-ZrO}_2$ as the support. Typically, the measured amount of $\text{Zn}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$ was dissolved in 100 mL deionized water, after which the pH of the solution was adjusted to 10 with 25 wt.% ammonia solution. Then, the calculated amount of $m\text{-ZrO}_2$ with a fixed surface loading of $10^3 \text{ m}^2 \text{ L}^{-1}$ was mixed with 100 mL of the above $\text{Zn}(\text{NO}_3)_2$ aqueous solution at 25°C . The resulting slurry was magnetically stirred for 3 h and filtrated. Then the recovered solid was dried at 80°C for 12 h and finally calcined at 400°C for 3 h in air to obtain the catalyst.

Structural characterizations

Powder X-ray diffraction (XRD) patterns with a 2θ range of $10\text{--}80^\circ$ were measured using a Philips X'Pert PROS diffractometer with a nickel-filtered Cu K α radiation source (0.15418 nm) operated with a voltage of 40 kV and a current of 50 mA, respectively. Transmission electron microscopy (TEM) and high-resolution transmission electron microscopy (HRTEM) images were obtained on Talos F200X and Titan Themis G2 60-300 microscopes, both operated at an accelerating voltage of 200 kV. Specific surface areas were determined using the Brunauer–Emmett–Teller (BET) method on a Micromeritics ASAP 2020 system after an adequate degassing of the samples at 573 K for 2 h in N_2 . X-ray photoelectron spectroscopy (XPS) measurements were carried out on an ESCALAB 250 high performance electron spectrometer using monochromated Al K α ($h\nu = 1486.7 \text{ eV}$) as the excitation source. The C 1s binding energy of adventitious carbon species was set to 284.8 eV to correct the likely charging effect. Sample compositions were analyzed by inductive coupled plasma atomic emission spectroscopy (ICP-AES) using an Opmita 7300DV spectrometer, during which

the sample was heated to 423 K in 1 mL hydrofluoric acid and then diluted with a determined volume of deionized water for the analysis.

ESR spectra were recorded at 140 K on a JEOL JESFA200 ESR spectrometer operating at 9.087 GHz with a microwave power of 0.998 mW, a modulation frequency of 100 kHz, and a modulation amplitude of 0.35 mT. The catalyst powders were placed in a high-pressure stainless-steel reactor tube equipped with two shut-off valves, allowing the sample to be isolated from ambient air. Following CO₂ or CO hydrogenation reaction under designated conditions, the reactor was cooled to 303 K in the reaction atmosphere. Then the tube was sealed and disconnected from the reactor and then transferred into an Ar-filled glovebox (Mikrouna Co. Ltd., China, with O₂ levels below 0.1 ppm) where 100 mg of the treated catalyst powders were carefully transferred into an ESR sample tube, which was subsequently sealed for ESR measurements.

Zn and Zr *K*-edge XAFS spectra were collected at the BL14W1 beamlines at the Shanghai Synchrotron Radiation Facility (SSRF) using Si (111) crystal monochromators. Prior to the measurements, the catalyst samples were compressed into thin sheets (1 cm in diameter) and sealed between Kapton films. The XAFS spectra were recorded at room temperature using a Bruker 5040 4-channel Silicon Drift Detector. Zn *K*-edge extended X-ray absorption fine structure (EXAFS) spectra were collected in fluorescence mode. The Zn and Zr *K*-edge XANES spectra of a specific sample exhibited negligible changes in the line-shape and peak position across repeated scans. Reference spectra of standard materials (Zn foil and ZnO) were obtained in transmission mode. Data reduction and analysis were performed using the Athena and Artemis software packages.

Catalytic tests

Catalytic performance in the CO_x hydrogenation reactions was evaluated on a high-pressure fixed-bed flow reactor custom-built by Xiamen Han De Engineering Co., Ltd. Typically, 1 g Zn₂₀Zr₈₀ catalyst with a grain size of 30–60 mesh (250–600 μm) or 0.3 g Zn₁/m-ZrO₂ catalyst diluted with 0.7 g of quartz sand was loaded in a Ti reactor with a 10 mm inner diameter and pretreated in flowing Ar (flow rate: 30 mL min⁻¹) at 773 K for Zn₂₀Zr₈₀ or 673 K for Zn₁/m-ZrO₂ for 2 h prior to reaction. After the reactor was cooled down to 303 K, the CO_x + H₂ flow (Total pressure: 3 MPa; H₂: CO₂ = 3:1 or H₂: CO = 2:1) containing Ar with a concentration of 3% as an internal standard for the calculation of CO_x conversion was introduced with a total flow rate of 30 mL min⁻¹. The composition of effluent gas reaction was analyzed with an online gas chromatograph (Ruimin GC2060, Shanghai) equipped with both a thermal conductivity detector (TCD) and a flame ionization detector (FID). And the selectivity was calculated on a molar carbon basis. The activity and selectivity of the catalytic reaction were calculated as shown in Eqs. (1)–(6) below, in which X_i represents conversion of substance i , S_i represents selectivity of substance i , x represents carbon chain length of C₂₊ components and n_i represents moles of substance i :

$$X_{CO_x}(\%) = \frac{n_{CO \text{ or } CO_2 \text{ produced}} + n_{CH_3OH \text{ produced}} + n_{CH_4 \text{ produced}} + 2n_{DME \text{ produced}} + \sum x n_{C_{2-4} \text{ produced}}}{n_{CO_x \text{ input}}} \times 100\% \quad (1)$$

$$S_{CH_3OH}(\%) = \frac{n_{CH_3OH \text{ produced}}}{n_{CO \text{ or } CO_2 \text{ produced}} + n_{CH_3OH \text{ produced}} + n_{CH_4 \text{ produced}} + 2n_{DME \text{ produced}} + \sum x n_{C_{2-4} \text{ produced}}} \times 100\% \quad (2)$$

$$S_{CO \text{ or } CO_2}(\%) = \frac{n_{CO \text{ or } CO_2 \text{ produced}}}{n_{CO \text{ or } CO_2 \text{ produced}} + n_{CH_3OH \text{ produced}} + n_{CH_4 \text{ produced}} + 2n_{DME \text{ produced}} + \sum x n_{C_{2-4} \text{ produced}}} \times 100\% \quad (3)$$

$$S_{CH_4}(\%) = \frac{n_{CH_4 \text{ produced}}}{n_{CO \text{ or } CO_2 \text{ produced}} + n_{CH_3OH \text{ produced}} + n_{CH_4 \text{ produced}} + 2n_{DME \text{ produced}} + \sum x n_{C_{2-4} \text{ produced}}} \times 100\% \quad (4)$$

$$S_{DME}(\%) = \frac{2n_{DME \text{ produced}}}{n_{CO \text{ or } CO_2 \text{ produced}} + n_{CH_3OH \text{ produced}} + n_{CH_4 \text{ produced}} + 2n_{DME \text{ produced}} + \sum x n_{C_{2-4} \text{ produced}}} \times 100\% \quad (5)$$

$$S_{C_{2-4}}(\%) = \frac{\sum x n_{C_{2-4} \text{ produced}}}{n_{CO \text{ or } CO_2 \text{ produced}} + n_{CH_3OH \text{ produced}} + n_{CH_4 \text{ produced}} + 2n_{DME \text{ produced}} + \sum x n_{C_{2-4} \text{ produced}}} \times 100\% \quad (6)$$

In situ and temporal DRIFTS and online MS measurements

In situ and temporal high-pressure DRIFTS measurements were carried out using a Nicolet IS50 FT-IR spectrometer equipped with an in situ high pressure DRIFTS reaction cell (Harrick Scientific Products, INC) and connected with a mass spectrometer (Hiden Analytical, YQ074159). DRIFTS spectra were collected using a MCT/A detector with 128 scans per spectrum and a resolution of 4 cm⁻¹. The catalyst sample was placed on the reaction cell stage and pretreated under flowing Ar (flow rate, 30 mL min⁻¹) at 773 K for 2 hours. After cooling to the desired reaction temperature, the background spectrum was recorded. The feed gas was then switched to a mixture of 3 MPa CO_x + H₂ (H₂: CO₂ = 3:1 or H₂: CO = 2:1) with a flow rate of 30 mL min⁻¹ and then the catalyst was heated to the target temperature at a rate of 10 K min⁻¹, held for 1 h, and subsequently cooled to 303 K before being purged with Ar. During this entire process, temporal in situ DRIFTS spectra were continuously recorded. The peak fitting processes of temporal in situ DRIFTS spectra were carried out using the “Peak Resolve” function of the “Omnic” software integrated with the infrared spectrometer. Typically, a set of original temporal in situ DRIFTS spectra within a fixed wavenumber range for a specific vibrational mode was loaded and automatically processed by selecting a “linear” baseline correction, a “Gaussian/Lorentzian” peak shape, a position constrained within a reasonable range and a fixed full width at half maximum (FWHM) of the peak of the same species to acquire the fitted spectra.

CH₃OH chemisorption measurements were also performed on the Nicolet IS50 FT-IR spectrometer. The catalyst sample on the reaction cell stage was firstly pretreated in Ar at 773 K for 2 h and then cooled to 303 K whose spectra were taken as the background spectra. Then the reactor cell was evacuated by a mechanical pump, filled with the CH₃OH vapor for 30 min, and evacuated again. Then the DRIFTS spectra were recorded.

Sodium formate (Sigma Aldrich) was used as the formate source and deposited on the catalyst with weight loadings of 1%, 2%, and 5% using the wetness impregnation method. The catalyst was dried at 353 K overnight. The IR background was measured using the catalyst only at 303 K with Ar (30 mL/min). After that, the catalyst with sodium formate was loaded into the DRIFTS cell and baked at 353 K with Ar for 1 hour. Then, the spectra were taken after the DRIFTS cell was cooled down to room temperature. By integrating the peak area of the formate species, the calibration curves were obtained.

Theoretical calculations

Stochastic surface walking global optimization. The Stochastic Surface Walking (SSW) algorithm incorporates an automated climbing mechanism that drives a system from a local minimum to a high-energy configuration along a randomly chosen direction, which is inherited from the bias-potential driven constrained-Broyden-dimer (BP-CBD) method originally developed for transition state searches. The SSW algorithm has proven effective in exploring complex potential energy surfaces (PES) and predicting stable structures. In each SSW step, a modified potential energy surface V_{mod} is employed to guide the system from the current minimum R_t^0 to a high energy configuration R_t^H . This is achieved by sequentially adding a series of bias Gaussian potentials v_n (n is the index of the bias potential,

$n = 1, 2, \dots, H$) along the corresponding direction N_t^n as shown in Eq. (7), where R_t denotes the coordinate vector of the structure, and V_{real} corresponds to the original, unmodified potential energy surface; R_t^n represents the n -th local minimum encountered along the climbing trajectory on the modified PES, which is progressively shaped by the addition of n Gaussian bias potentials. Once the system reaches the high-energy configuration R_t^H , all bias potentials are removed, and a local geometry optimization is performed to relax the structure to a nearby minimum. At the end of each SSW step, a Metropolis Monte Carlo scheme is employed to determine whether the newly obtained minimum is accepted or rejected.

$$V_{mod} = V_{real} + \sum_{n=1}^{NG} V_n = V_{real} + \sum_{n=1}^{NG} w_n \cdot \exp[-((R_t - R_t^n) \cdot N_t^n)^2 / (2 \times ds^2)] \quad (7)$$

Neural Network potential energy surface. In this work, we utilized the NN scheme that firstly introduced by Behler and Parrinello for constructing the global NN PES. The total energy E^{tot} of a structure is decomposed and described as a linear combination of atomic energy E_i from the output of NN, where six power-type structure descriptors (PTSDs), namely S^1 to S^6 , are utilized as the input of the network. These descriptors describe the geometrical environment of atom in the following mathematic forms:

$$E^{tot} = \sum_i E_i \quad (8)$$

$$f_c(R_{ij}) = \begin{cases} 0.5 \times \tanh^3 \left[1 - \frac{r_{ij}}{r_c} \right], & \text{for } r_{ij} \leq r_c \\ 0, & \text{for } r_{ij} > r_c \end{cases} \quad (9)$$

$$R^n(r_{ij}) = r_{ij}^n \cdot f_c(r_{ij}) \quad (10)$$

$$S_i^1 = \sum_{j \neq i} R^n(r_{ij}) \quad (11)$$

$$S_i^2 = \left[\sum_{m=-L}^L \left| \sum_{j \neq i} GU_2 \right|^2 \right]^{\frac{1}{2}} = \left[\sum_{m=-L}^L \left| \sum_{j \neq i} R^n(r_{ij}) Y_{Lm}(r_{ij}) \right|^2 \right]^{\frac{1}{2}} \quad (12)$$

$$S_i^3 = 2^{1-\zeta} \sum GU_3 = 2^{1-\zeta} \sum_{j,k \neq i} (1 + \lambda \cos \theta_{ijk})^\zeta \cdot R^n(r_{ij}) \cdot R^m(r_{ik}) \quad (13)$$

$$S_i^4 = 2^{1-\zeta} \sum GU_4 = 2^{1-\zeta} \sum_{j,k \neq i} (1 + \lambda \cos \theta_{ijk})^\zeta \cdot R^n(r_{ij}) \cdot R^m(r_{ik}) \cdot R^p(r_{ik}) \quad (14)$$

$$S_i^5 = \left[\sum_{m=-L}^L \left| \sum_{j,k \neq i} GU_5 \right|^2 \right]^{\frac{1}{2}} = \left[\sum_{m=-L}^L \left| \sum_{j,k \neq i} R^n(r_{ij}) \cdot R^m(r_{ik}) \cdot R^p(r_{ik}) \cdot (Y_{Lm}(r_{ij}) + Y_{Lm}(r_{ik})) \right|^2 \right]^{\frac{1}{2}} \quad (15)$$

$$S_i^6 = 2^{1-\zeta} \sum GU_6 = 2^{1-\zeta} \sum_{j,k,l \neq i} (1 + \lambda \cos \theta_{ijk})^\zeta \cdot R^n(r_{ij}) \cdot R^m(r_{ik}) \cdot R^p(r_{il}) \quad (16)$$

The dataset for training the neural network potential energy surface (NN PES) is constructed through an iterative process. The initial dataset is generated from density functional theory (DFT)-based SSW simulations, while subsequent data are obtained from SSW explorations driven by the evolving NN PES. These SSW simulations cover a wide range of compositions and morphologies—including bulk, layered, and cluster structures—with varying atomic compositions and unit cell sizes. In total, the simulations sampled over 10^7 structures on the PES, from which 58,883 representative structures were selected to form the final global dataset, all of which were recalculated using high-accuracy DFT. To ensure high accuracy of the PES, we employed a comprehensive set of PTSDs, comprising 493 descriptors per element: 282 two-body, 171 three-body, and 40 four-body terms. Accordingly, a large neural network architecture was adopted, consisting of two hidden layers with a 100-80-80-6 structure.

SSW-NN calculations. Traditional density functional theory (DFT) calculations are often impractical for the global optimization of complex systems due to their prohibitively high computational cost. To overcome this limitation, we employed SSW global optimization based on NN PES to resolve the structure of the ZnZrO slab model, achieving performance 3 to 4 orders of magnitude faster than conventional DFT. The application of the SSW-NN method follows six main steps: (1) Generating a global dataset from the SSW trajectories and computing reference energies and forces using DFT; (2) Training the NN PES on this dataset; (3) Benchmarking the NN predictions against DFT results for selected structures from SSW trajectories, and augmenting the dataset with new configurations for retraining; (4) Iterating steps (1) through (3) until the PES deviation is sufficiently small (typically below 10 meV/atom); (5) Conducting SSW-based global optimization using the converged NN PES for the target system; (6) Recalculating the energies of key structures using high-accuracy DFT. In this study, the NN PES developed for the ZnZrO system achieves good agreement with DFT, with energy and force deviations of 3.696 meV/atom and 0.117 eV/Å, respectively.

The Gibbs free energy. For solids, the free energy is approximated by its DFT total energy. For molecules, the free energy is calculated as Eq. (17), including the DFT total energy $[X]$, the zero-point-energy (ZPE) and the thermodynamic correction terms. The enthalpy changes and entropies for CO, CO₂, H₂, and H₂O under reaction and standard conditions, as well as for CH₃OH under the standard condition, are obtained from the NIST-JANAF Thermochemical Tables⁶⁰ and the enthalpy change and entropy of CH₃OH under the reaction condition are calculated from DFT⁶¹ (Supplementary Table 9).

$$G[X](p, T) = E[X] + ZPE[X] + [H[X](p^0, T) - H[X](p^0, 0K) - TS[X](p^0, T) + kBT \ln p / p^0] \quad (17)$$

Surfaces search. We employed Stochastic Surface Walking global optimization^{62–64} with global neural network (G-NN) potential method (SSW-NN)^{65,66} to characterize the different Zn concentration doping on the *t*-ZrO₂ (101) surface structures. The global minimums (GM) of active site structures were determined by the SSW-NN method, as implemented in the large-scale atomic simulation with neural network potential (LASP) code⁵⁴. The Zn–Zr–O PES describes the global neural network (G-NN) potential that was obtained by self-learning the dataset produced from SSW global optimization. The G-NN potential is available from the LASP project (www.lasphub.com). More details on SSW-NN and the Zn–Zr–O G-NN potential are provided in the Supporting Information.

In the SSW-NN simulation, the *t*-ZrO₂ (101) surface is modeled by a five-layer slab, where the bottom three layers were fixed at the bulk-truncated position and the top two layers were allowed to relax.

Considering that the *t*-ZrO₂ (101) surface supercell contains 12 Zr atoms, the doping we considered corresponds to 8.3% (1 Zr substituted on surface), 16.6% (2 Zr substituted on surface), and 25.0% (3 Zr substituted on surface) Zn atoms on the surface. The supercell had a dimension of 12.8 Å × 10.9 Å × 30.0 Å. And the relative stability is given by Eq. (18):

$$\Delta E_f = [E(\text{Zn}_3\text{ZrO}_2) + 2 \times E(\text{slab} - \text{ZrO}_2) - 3 \times E(\text{Zn}_1\text{ZrO}_2)]/3 \quad (18)$$

DFT calculations. The reaction pathways on the ZnZrO active site were calculated using first-principles DFT as implemented in the Vienna Ab initio Simulation Package (VASP)⁶⁷. The exchange–correlation function utilized was GGA-PBE⁶⁸. The electron–ion interaction was represented by the projector-augmented-wave pseudopotential^{69,70}, and the kinetics energy cutoff utilized was 450 eV. The first Brillouin zone k-point sampling utilizes only the gamma-point since the supercell was rather large, which was shown to provide converged energetics. The energy and force criterion for convergence of the electron density and structure optimization were set at 10^{−5} eV and 0.05 eV/Å, respectively. All of the reaction TSs were determined using the constrained Broyden dimer (CBD) method^{64,71}, and the TS was further confirmed by the vibrational frequency analysis and the geometry extrapolation to the neighboring minima.

Microkinetic simulation. Microkinetic simulation was performed to evaluate the theoretical apparent activation energy. In the simulation, the pressure of CO₂ and H₂ were fixed at the reaction condition (3 MPa, H₂: CO₂ = 3) to simulate a fluidized-bed catalytic reactor during the whole process. The selected temperatures were 548, 558 and 573 K.

Data availability

The data supporting the findings of the study are available within the paper and its Supplementary Information. Source data are provided with this paper.

Code availability

The software code of LASP and Zn-Zr-O G-NN potential used within the article are available from the corresponding author (C.S.) upon request or from the website <http://www.lasphub.com>.

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Acknowledgements

This work was financially supported by the National Key R&D Program of MOST (2022YFA1504601, 2018YFA0208600), the Strategic Priority Research Program of the Chinese Academy of Sciences (XDB0450102), the National Natural Science Foundation of China (92145302, 91745202, 22122301, 92061112, 22033003, 22250710677), the Fundamental Research Funds for the Central Universities (20720220008, 20720220011), the Changjiang Scholars Program of Ministry of Education of China, the University of Science and Technology of China (KY2060000176), and the Tencent Foundation for XPLOER PRIZE. This work was partially carried out at the Instruments Center for Physical Science, University of Science and Technology of China.

Author contributions

J.D. and Y.P. contributed equally to the work. W.H. conceived and supervised the project, oversaw all data analysis and discussion. C.S. supervised the theoretical simulations. project J.D. performed the experiments and analyzed the data. Y.P. and C.S. performed the theoretical simulations and analyzed the data. W.X., D.W. and Q.N. assisted with the experiments. Z.X. and Z.J. assisted with the EXAFS data analysis. Z.L. assisted with the DFT calculation. W.H., J.D., C.S. and Y.P. prepared the manuscript. All the authors discussed the results and commented on the manuscripts.

Competing interests

The authors declare no competing interests.

Additional information

Supplementary information The online version contains supplementary material available at <https://doi.org/10.1038/s41467-025-59996-5>.

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Peer review information *Nature Communications* thanks Bert Wec-khuysen and the other, anonymous, reviewer(s) for their contribution to the peer review of this work. A peer review file is available.

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